

The SCX Inquisition:

A Complete Audit of Machine Learning —
What Every Algorithm Lacks and Why

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SCX

June 2026

摘要

Machine learning’s 70-year history has produced thousands of algorithms. Not one declares its M parameter —the minimum number of independent experts needed to certify output quality at error rate ε . Not one provides a formal quality guarantee backed by multi-expert consensus. We audit 27 major ML algorithms against the SCX framework’s five theorems (multi-expert detection, self-audit prohibition, noise-signal unidentifiability, reviewer complicity, and Cercis scoring). Every algorithm is found **UNDECLARED** of at least one structural deficiency. The best classical algorithm (Random Forest, Random Forest) is deemed **VERIFIED** —it accidentally implements Theorem 1 through bootstrap + random subspace sampling. The worst (GAN, RL, Self-Supervised Learning) are **UNDECLARED** of $M = 1$ or self-audit —epistemically equivalent to no audit at all. We rank all 27 algorithms by the CERCIS Score $S = Q + \eta N$, where Q measures formal quality guarantees and N measures empirical novelty. The verdict: the 70-year history of ML is the history of algorithms that work —until they don’t —with no way to know when or why. SCX provides the answer: declare M . If M is high and verified, trust. If M is low or undeclared, don’t.

Keywords: SCX audit, machine learning, multi-expert consensus, quality guarantee, M parameter, Cercis Score, Theorem 1–5, Machine Learning Audit, Quality Assurance,

1 Prologue — The Indictment Prologue — The Indictment

Machine learning is 70 years old. From Rosenblatt’s perceptron (1958) to today’s trillion-parameter transformers, the field has produced an extraordinary diversity of algorithms: linear models, kernel machines, decision trees, ensembles, deep networks, generative models, reinforcement learners, self-supervised systems. Thousands of papers. Millions of deployments. Billions in investment.

Not one declares its M parameter.

The M parameter is the minimum number of independently-trained experts whose consensus is required to certify output quality at error rate ε . More precisely, for a learning system producing predictions $\hat{y}$ on inputs x , M is the smallest integer such that:

$$\mathbb{P}(\text{all } M \text{ experts miss error of magnitude } \Delta \mid \text{error exists}) \leq \varepsilon. \quad (1)$$

This is the most fundamental epistemic quantity in any prediction system: how many independent verifiers would need to agree before you can trust the output? Every scientific claim requires verifiability (Theorem 2 of SCX); every ML prediction is a scientific claim about an unseen datum. Yet no algorithm specifies M . No paper declares it. No benchmark measures it.

This is not a minor oversight. It is the structural absence of audit from the entire field. ML has optimized for accuracy, speed, interpretability, fairness, robustness —everything except verifiability. The result is algorithms that achieve state-of-the-art performance on benchmarks, deployed in production, making decisions affecting human lives —with **zero formal guarantee** that their errors would be detected.

Consider the implications:

- A medical diagnosis model with 95% test accuracy can be wrong 5% of the time. How do you know which 5% are wrong? You don’t. The model doesn’t either.
- A self-driving car perception system detects pedestrians with 99.9% recall. The remaining 0.1% are undetected pedestrians. Which ones? Nobody knows.
- A credit scoring model approves loans. Some approvals default. The model has no mechanism to flag the risky ones *before* default occurs.

In each case, the *average* performance is good, but the *individual* prediction quality is unknown. This is the audit gap: the distance between aggregate metrics and instance-level verification. SCX closes this gap by requiring multiple independent experts to agree on each prediction.

1.1 The 70-Year Audit Gap The 70-Year Audit Gap

The history of ML can be periodized by its relationship to audit:

- Era 1: Statistical Learning (1958–1995):** Algorithms optimize loss functions. No concept of verification beyond test set accuracy. M is never mentioned.
- Era 2: Ensemble Methods (1995–2012):** Multiple models improve performance. The intuition of consensus emerges, but without formalization. M exists implicitly but is never declared.
- Era 3: Deep Learning (2012–2020):** Depth compensates for everything —noise, bias, variance, distribution shift. Architecture becomes the answer to every question. M is forgotten entirely.
- Era 4: Foundation Models (2020–present):** Single colossal models replace ensembles. Scale replaces diversity. $M = 1$ at unprecedented parameter counts.

Throughout these 70 years, the field optimized for **prediction quality** while ignoring **verification quality**. The two are not the same. A model can predict perfectly on average while providing zero information about which predictions are reliable. SCX is the first framework to demand both.

2 The Audit Framework The Audit Framework

2.1 SCX Core Assumptions (A1–A5) SCX Core Assumptions

The SCX framework rests on five axioms that define what it means for a learning system to be auditable. Every ML algorithm is measured against these axioms.

Assumption 2.1 (Independent Expert Training —A1 Independent Training). Each expert $f_m \in \mathcal{F} = \{f_1, \dots, f_M\}$ is trained on independent data subsets $\mathcal{D}_m \subset \mathcal{D}$ or with independent random initializations and stochastic training trajectories. Formally, for any $m \neq m'$, the training procedures for f_m and $f_{m'}$ are statistically independent: $I(\mathcal{D}_m; \mathcal{D}_{m'}) = 0$ and $I(\theta_m^{(0)}; \theta_{m'}^{(0)}) = 0$, where $\theta_m^{(0)}$ denotes initial parameters.

Assumption 2.2 (Conditional Expert Independence —A2 Conditional Independence). Given the true label y , expert predictions $\hat{y}_1, \dots, \hat{y}_M$ are conditionally independent:

$$\mathbb{P}(\hat{y}_m = a, \hat{y}_{m'} = b \mid y) = \mathbb{P}(\hat{y}_m = a \mid y) \cdot \mathbb{P}(\hat{y}_{m'} = b \mid y), \quad \forall m \neq m'.$$

In practice, exact independence is impossible —experts trained on data from the same distribution have correlated errors. We define the **effective multiplicity**:

$$M_{\text{eff}} = \frac{M}{1 + \bar{\rho}}, \tag{2}$$

where $\bar{\rho} = \frac{2}{M(M-1)} \sum_{m < m'} (\mathbf{1}[\hat{y}_m \neq y], \mathbf{1}[\hat{y}_{m'} \neq y])$ is the average inter-expert error correlation. When $\bar{\rho} = 0$ (perfect independence), $M_{\text{eff}} = M$. When $\bar{\rho} \rightarrow \infty$ (total correlation), $M_{\text{eff}} \rightarrow 1$.

Assumption 2.3 (Bounded Expert Competence — A3 Bounded Competence). Each expert achieves accuracy strictly better than random: for K -class problems,

$$\mathbb{P}(\hat{y}_m = y \mid y) > \frac{1}{K}, \quad \forall m \in [M].$$

Equivalently, each expert's error probability is bounded above by $1 - 1/K$.

Assumption 2.4 (Detectable Error Margin — A4 Detectable Error Margin). When a prediction error of magnitude $\Delta > 0$ occurs, at least one expert detects it with probability $p_d \geq 1/2 + \gamma$ for some $\gamma > 0$. Formally, defining $Z_m = \mathbf{1}[\|\hat{y}_m - y\| > \Delta]$:

$$\mathbb{E}[Z_m \mid \text{error exists}] \geq \frac{1}{2} + \gamma.$$

This ensures the majority of experts are correct — a necessary condition for consensus to be informative.

Assumption 2.5 (Bounded State Distribution — A5 Bounded State Distribution). The state distribution ρ_s over which predictions are made is stationary and has bounded support: $\text{supp}(\rho_s) \subseteq \mathcal{X}$ is compact. Distribution shift $\rho_s \rightarrow \rho_t$ can be detected via SPRING gating but not compensated without re-audit.

2.2 SCX Theorems as Audit Criteria SCX Theorems as Audit Criteria

Theorem 1 (Multi-Expert Detection Bound Multi-Expert Detection Bound). Under A1–A4, for M independent experts with effective multiplicity M_{eff} , the probability of missing an error of magnitude Δ satisfies:

$$\mathbb{P}(\text{all } M \text{ experts miss error} \mid \text{error}) \leq \exp(-2M_{\text{eff}}\Delta^2). \quad (3)$$

Proof sketch: Each expert m detects the error independently with probability $p_m \geq 1/2 + \gamma$. The worst case is when all $p_m = 1/2 + \gamma$ (minimum competence). The probability all miss is $\prod_m (1 - p_m) \leq (1/2 - \gamma)^M$. By Hoeffding's inequality applied to the Bernoulli indicators Z_m , the probability that the sample mean $\bar{Z} = \frac{1}{M} \sum_m Z_m$ deviates from its expectation by more than γ is bounded by $\exp(-2M\gamma^2)$. Setting $\gamma = \Delta$ and using M_{eff} for correlated experts yields the bound. $\square$

Corollary 1.1: For $M = 1$, $\mathbb{P}(\text{miss}) \leq \exp(-2\Delta^2)$, which equals $e^{-2} \approx 0.135$ when $\Delta = 1$. A single expert provides at most 86.5% error detection confidence — unacceptable

for any safety-critical application.

Corollary 1.2: To achieve $\mathbb{P}(\text{miss}) \leq \varepsilon$, the required number of experts is:

$$M \geq \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot (1 + \bar{\rho}). \quad (4)$$

For $\varepsilon = 0.05$, $\Delta = 0.5$, $\bar{\rho} = 0.2$: $M \geq 7.2 \Rightarrow M \geq 8$.

Theorem 2 (Self-Audit = No Audit Self-Audit = No Audit). *A learning system that generates its own labels $\hat{y}$ and evaluates itself against those labels provides zero information about true quality. Formally:*

$$I(\text{self-audit outcome}; \text{true error}) = 0 \quad (5)$$

when the audit signal is a deterministic function of the prediction.

Proof: Let the system produce prediction $\hat{y} = f(x)$ and self-audit signal $a = g(f(x), x)$ for some function g . The self-audit signal is a deterministic function of x . The true error $e = \mathbf{1}[\hat{y} \neq y]$ depends on y . Since y is independent of x given the data-generating process, and a depends only on x , we have $I(a; e) = 0$ by the data processing inequality, unless g has access to y (which would make it not a self-audit). $\square$

Corollary 2.1: Any algorithm that uses training accuracy, validation accuracy on a fixed hold-out set, or self-generated confidence scores as its only quality signal is epistemically equivalent to having no quality signal. External, independent auditors are required.

Corollary 2.2: Cross-validation with K folds partially escapes self-audit —each fold’s model is evaluated on data it never saw during training. However, the K models share the same architecture and hyperparameters, so $M_{\text{eff}} \approx K/(1 + \bar{\rho}_{\text{arch}})$ where $\bar{\rho}_{\text{arch}}$ is architectural correlation. $K = 5$ with $\bar{\rho}_{\text{arch}} = 0.3$ gives $M_{\text{eff}} \approx 3.8$.

Theorem 3 (Noise-Signal Unidentifiability Noise-Signal Unidentifiability). *For any architecture using L layers of transformations T_ℓ , it is impossible to distinguish whether layer ℓ corrects genuine modeling error or memorizes training noise, unless each layer’s contribution is independently audited by $M > 1$ experts.*

$$\forall \ell \in [L], \nexists \text{ detector } D_\ell : \mathcal{X} \rightarrow \{0, 1\} \text{ s.t. } D_\ell(T_\ell(\cdots T_1(x))) = \mathbf{1}[\text{noise}], \text{ without } M > 1. \quad (6)$$

Proof sketch: Let $h_\ell = T_\ell(h_{\ell-1})$ be the output of layer ℓ . Suppose $h_\ell = s + n$ where s is signal and n is noise. A single model cannot distinguish s from n because it has no independent reference for what the “correct” h_ℓ should be. Any attempt to estimate s from h_ℓ itself (e.g., via reconstruction loss) falls under Theorem 2 (self-audit). With M independent models, each producing $h_\ell^{(m)}$, the consensus $h_\ell^* = \text{median}_m(h_\ell^{(m)})$ estimates signal, and the deviation $\|h_\ell^{(m)} - h_\ell^*\|$ estimates noise. $\square$

Corollary 3.1: Adding layers to a deep network without independent per-layer audit

is epistemically equivalent to adding degrees of freedom to memorize noise. Depth can improve performance and simultaneously degrade verifiability —and you cannot tell which effect dominates.

Theorem Situs-1 (Encoding Quality Encoding Quality). *A SITUS encoding $\phi : \mathcal{X} \rightarrow \mathcal{Z}$ is quality-certified if and only if it has bounded Lipschitz constant $\text{Lip}(\phi) \leq L_\phi$ and the reconstruction error satisfies $\|x - \psi(\phi(x))\| \leq \varepsilon_\phi$ for decoder ψ . Encodings without Lipschitz bounds are unverifiable.*

$$Q_{\text{SITUS}}(\phi) = (1 - \varepsilon_\phi) \cdot \exp(-L_\phi \cdot \sigma_x^2), \quad (7)$$

where σ_x^2 is the input variance. High Lipschitz constant (sensitive encoding) or high reconstruction error $\rightarrow$ low encoding quality.

Theorem Spring-1 (Permanent Memory Permanent Memory). *SPRING memory M_t is monotonic non-decreasing: $M_t \supseteq M_{t-1}$ for all t . Any memory mechanism that permits voluntary forgetting —graded decay of stored evidence —is a bug, not a feature. The detection probability at time t satisfies $\mathbb{P}(D_t) \geq \mathbb{P}(D_{t-1})$ and $\lim_{t \rightarrow \infty} \mathbb{P}(D_t \mid \text{fraud}) = 1$.*

2.3 The Five SCX Theorems —Formal Statement The Five SCX Theorems — Formal Statement

For completeness, we restate all five theorems in their most general form. These are the standards against which every algorithm is judged.

Theorem 2.6 (Multi-Expert Error Detection —Theorem 1 Multi-Expert Error Detection). *Let $\mathcal{F} = \{f_1, \dots, f_M\}$ be M experts satisfying A1–A4. Define $M_{\text{eff}} = M/(1 + \bar{\rho})$ where $\bar{\rho}$ is the average inter-expert error correlation. For any error of magnitude $\Delta > 0$, the probability that all M experts fail to detect it satisfies:*

$$\mathbb{P}(\cap_{m=1}^M \{\|\hat{y}_m - y\| \leq \Delta\}) \leq \exp(-2M_{\text{eff}}\Delta^2).$$

Consequences: (1) $M = 1 \implies$ no formal guarantee; (2) To achieve $\mathbb{P}(\text{miss}) \leq \varepsilon$, require $M \geq \ln(1/\varepsilon)/(2\Delta^2) \cdot (1 + \bar{\rho})$; (3) The bound is tight —Hoeffding’s inequality is sharp for bounded random variables.

Theorem 2.7 (Self-Audit Prohibition —Theorem 2 Self-Audit Prohibition). *Let $\mathcal{S}$ be a learning system that generates quality labels $\hat{q}$ from its own predictions $\hat{y} = f(x)$: $\hat{q} = g(f(x), x)$ for some function g . Then the mutual information between the self-generated quality signal and the true error $e = \mathbf{1}[\hat{y} \neq y]$ is zero:*

$$I(\hat{q}; e) = 0,$$

unless g has access to y , in which case it is not a self-audit. Consequence: Any system that evaluates itself using only its own outputs provides zero information about its true error rate.

Theorem 2.8 (Noise-Signal Unidentifiability —Theorem 3 Noise-Signal Unidentifiability). *For any composition of L transformations $T_L \circ \dots \circ T_1$, it is impossible to determine whether transformation T_ℓ corrects genuine error or memorizes noise, using only the outputs of the composed function. Formally: $\forall \ell$, there exists no detector D_ℓ depending only on $(T_\ell \circ \dots \circ T_1)(x)$ that can distinguish between $\text{CorrectionRatio}_\ell > \tau$ (signal) and $\text{CorrectionRatio}_\ell \leq \tau$ (noise) with probability $> 1/2 + \delta$ for any $\delta > 0$, unless $M > 1$ independent copies of the composition are available.*

Theorem 2.9 (Situs Encoding Quality —Theorem Situs-1 SitusEncoding Quality). *An encoding $\phi : \mathcal{X} \rightarrow \mathcal{Z}$ with decoder $\psi : \mathcal{Z} \rightarrow \mathcal{X}$ is quality-certified iff: (i) $\text{Lip}(\phi) \leq L_\phi$ for some known $L_\phi < \infty$, (ii) $\mathbb{E}_{x \sim \rho}[\|x - \psi(\phi(x))\|] \leq \varepsilon_\phi$. The encoding quality score is $Q_{\text{SITUS}}(\phi) = (1 - \varepsilon_\phi) \cdot \exp(-L_\phi \cdot \sigma_x^2)$. Encodings without Lipschitz bounds or reconstruction guarantees are unverifiable.*

Theorem 2.10 (Spring Permanent Memory —Theorem Spring-1 SpringPermanent Memory). *Spring memory M_t is monotonic: $M_t \supseteq M_{t-1}$ for all $t \geq 1$. The detection probability is non-decreasing: $\mathbb{P}(D_t \mid \text{fraud}) \geq \mathbb{P}(D_{t-1} \mid \text{fraud})$. As $t \rightarrow \infty$ with unbounded verifier growth (Assumption A8 from the Science Audit paper), $\lim_{t \rightarrow \infty} \mathbb{P}(D_t \mid \text{fraud}) = 1$. Any mechanism that permits voluntary forgetting (graded memory decay, bounded capacity, context window limits) violates this theorem.*

These five theorems define the audit criteria. Every algorithm is measured by how well it satisfies them —structurally, not empirically. An algorithm either has $M > 1$ independent experts or it doesn't. It either has permanent memory or it doesn't. It either avoids self-audit or it doesn't. There is no “approximately satisfies Theorem 2.”

2.4 The Cercis Score —Quantifying Algorithmic Quality Cercis Score

Definition 2.11 (CERCIS Score for ML Algorithms Cercis Score for ML Algorithms). *For an ML algorithm A , the CERCIS Score is:*

$$\text{CERCIS}(A) = Q(A) + \eta \cdot N(A), \quad (8)$$

where:

- $Q(A) \in [0, 1]$ is the **Quality Guarantee (Quality Assurance)**: the fraction of SCX theorems the algorithm implicitly or explicitly satisfies. Specifically:

$$Q(A) = \frac{1}{5} \left[q_1(A) + q_2(A) + q_3(A) + q_4(A) + q_5(A) \right], \quad (9)$$

where $q_i(A) \in [0, 1]$ is the degree to which algorithm A satisfies Theorem i (Theorems 1–3, Situs-1, Spring-1). $q_i = 1$ means the theorem is formally satisfied with proof; $q_i = 0$ means violated.

- $N(A) \in [0, 1]$ is the **Empirical Novelty (Empirical Novelty)**: normalized performance across standard benchmarks.
- $\eta = 0.2$ is the **epistemic discount factor (epistemic discount factor)**: empirical success without formal guarantee is worth at most 20%.

Critical note: If $Q(A) = 0$, then $\text{CERCIS}(A) \leq 0.2$ regardless of empirical performance. An algorithm with 99% benchmark accuracy and no quality guarantee is epistemically worthless.

The CERCIS Score operationalizes a fundamental principle: **novelty without verifiability is noise**. An algorithm that works well empirically but cannot tell you *when* it works well is epistemically equivalent to a random guess with lucky initialization.

2.5 Audit Verdict Categories Audit Verdict Categories

Every algorithm is classified into one of four categories:

- **VERIFIED VERIFIED (VERIFIED)**: $M > 1$, independent experts, detectable errors. The algorithm structurally satisfies Theorems 1–3, Situs-1, or Spring-1. Typical $\text{CERCIS}(A) \geq 0.70$.
- **PARTIAL PARTIAL (PARTIAL)**: $M > 1$ but with structural limitations. Approximates SCX guarantees but lacks formal proof or has correlation issues. Typical $0.45 \leq \text{CERCIS}(A) < 0.70$.
- **INCOMPLETE INCOMPLETE (INCOMPLETE)**: Violates a core SCX assumption ($M = 1$, self-audit, sequential dependency). Outputs are structurally unverifiable. Typical $0.20 \leq \text{CERCIS}(A) < 0.45$.
- **UNDECLARED UNDECLARED (UNDECLARED)**: Never considered audit at all. M is undeclared, zero, or structurally undefined. Typical $\text{CERCIS}(A) < 0.20$.

This paper is a structural audit, not a celebration. We do not ask “does this algorithm work?” —many work, empirically, most of the time. We ask: “can this algorithm tell you when it stops working?” The answer, for nearly all, is no.

3 Part I: Classical ML (pre-2012) —The Age of Innocence

4 Part I: Classical Machine Learning — The Age of Innocence

The classical era produced algorithms that are mathematically elegant, computationally tractable, and well-understood. But mathematical elegance is not epistemological hygiene. None of these algorithms considered audit. They were built to predict, not to verify.

4.1 1. Linear / Logistic Regression Linear / Logistic Regression

Verdict: **UNDECLARED** — $M = 1$, Zero Audit Structure

Why Popular: Mathematically tractable. Linear regression solves $\min_w \|Xw - y\|_2^2$ with closed form $\hat{w} = (X^\top X)^{-1} X^\top y$. Logistic regression adds the sigmoid link: $p(y = 1|x) = \sigma(w^\top x) = 1/(1 + e^{-w^\top x})$, solved via convex optimization. Both provide interpretable coefficients, confidence intervals (under Gaussian assumptions), and p -values. The foundation of statistical learning since Legendre (1805) and Gauss (1809).

Failure Mode: **Cannot detect its own errors.** The model produces $\hat{y} = w^\top x$ with residual $r = y - \hat{y}$. The residual is computable only *after* observing y —by which point audit is irrelevant. Before y is known, the model provides a point estimate. The confidence interval $\hat{y} \pm t_{\alpha/2} \cdot \hat{\sigma} \sqrt{1 + x^\top (X^\top X)^{-1} x}$ assumes the model is correctly specified —an assumption the model cannot verify.

SCX Diagnosis: $M = 1$. A single weight vector produces a single prediction. No second opinion. Theorem 1: $\mathbb{P}(\text{miss}) \leq \exp(-2\Delta^2)$ with $M_{\text{eff}} = 1$. For $\Delta = 1$ (one standard deviation error), detection probability is at most $1 - e^{-2} \approx 0.865$. Translating to human terms: the model admits a 13.5% chance of missing a one-sigma error.

Mathematical Audit: The quality guarantee is:

$$Q_{\text{LR}} = \frac{1}{5} \left[\underbrace{0}_{\text{Thm 1: } M=1} + \underbrace{0}_{\text{Thm 2: self-audit}} + \underbrace{0}_{\text{Thm 3: no depth}} + \underbrace{0}_{\text{Situs-1: no encoding}} + \underbrace{0}_{\text{Spring-1: no memory}} \right] = 0. \quad (10)$$

With $N = 0.1$ (widely used, but simple): $\text{CERCIS} = 0 + 0.2 \cdot 0.1 = 0.02$ —effectively zero.

What SCX Provides: Multi-expert linear consensus. Train M linear models on M independent bootstrap samples. The YAJIE consensus $\hat{y}_{\text{YAJIE}} = \text{weighted_median}(\hat{y}_1, \dots, \hat{y}_M)$ with weights $\omega_m = 1/\hat{\sigma}_m^2$ (inverse variance weighting). Theorem 1 applies: $\mathbb{P}(\text{miss}) \leq \exp(-2M_{\text{eff}}\Delta^2)$. SPRING permanently records every $(x, \hat{y}_m, y)$ triple for retrospective audit. The multi-expert linear regression would have $Q = 0.6$ and $\text{CERCIS} = 0.62$.

4.2 2. k-Nearest Neighbors (k-NN) k-Nearest Neighbors

Verdict: **UNDECLARED** —Lazy Audit, No State Quality Guarantee

Why Popular: Beautifully simple: store training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, predict $\hat{y}(x) = \text{mode}\{y_i : x_i \in \mathcal{N}_k(x)\}$ where $\mathcal{N}_k(x)$ is the k nearest neighbors under distance $d(x, x_i)$. No training phase. Non-parametric —no assumptions about $p(y|x)$. The decision boundary adapts to local data density. Intuitive: “your label is what your neighbors say.”

Failure Mode: **Curse of dimensionality = state space explosion.** In d dimensions, to maintain constant neighbor density δ , the required sample size n grows as $O((1/\delta)^d)$. In high dimensions, all pairwise distances converge: $\max_{i,j} d(x_i, x_j) / \min_{i,j} d(x_i, x_j) \rightarrow 1$ as $d \rightarrow \infty$. The concept of “nearest” loses meaning. The k neighbors define an implicit state partition, but there is no quality guarantee on that partition.

SCX Diagnosis: k-NN has implicit state clustering (k neighbors define a local state region), resembling SITUS encoding. But the encoding is implicit, lazy, and unverifiable. The Lipschitz constant of the encoding is $\text{Lip}(\phi_{\text{kNN}}) = \max_{x,x'} \|\phi(x) - \phi(x')\| / \|x - x'\|$, which can be arbitrarily large —a small perturbation in x can flip the neighbor set, producing discontinuous prediction changes.

Mathematical Audit: The k neighbors are **not** independent experts —they are training points from the same dataset. The “vote” is not a YAJIE consensus because: (1) neighbors are correlated (nearby points share similar features and labels), (2) neighbors are not independently-trained models with distinct architectures, (3) $M_{\text{eff}} \ll k$ due to spatial correlation. With $\bar{\rho} \approx 0.7$ (typical for spatial data), $M_{\text{eff}} = k/1.7$. For $k = 5$, $M_{\text{eff}} \approx 2.9$.

What SCX Provides: SITUS encoding with explicit Lipschitz bound $\text{Lip}(\phi) \leq L$. Train M independent k-NN classifiers on disjoint data splits. YAJIE consensus across M k-NN outputs with effective multiplicity M_{eff} . SPRING permanently records which neighbors voted for each prediction. CERCIS Score for state partition quality: $Q_{\text{state}} = \exp(-\text{Var}[\hat{p}(y|x)])$ measuring prediction stability.

4.3 3. Naive Bayes Naive Bayes

Verdict: **UNDECLARED** —Extreme Independence Assumption, Never Verified

Why Popular: Fast, simple, surprisingly effective. By Bayes’ rule with the “naive” conditional independence assumption:

$$P(y \mid x_1, \dots, x_d) \propto P(y) \prod_{i=1}^d P(x_i \mid y).$$

This reduces the parameter count from $O(2^d)$ to $O(d)$ —exponential simplification. Works well for text classification (spam detection, sentiment analysis) where bag-of-words features are approximately conditionally independent. Computationally efficient: $O(nd)$ training, $O(d)$ inference.

Failure Mode: The independence assumption $P(x_i, x_j|y) = P(x_i|y)P(x_j|y)$ is false in most real-world data. Features are correlated. Naive Bayes has no mechanism to (a) measure how badly the assumption is violated, (b) correct for violations, or (c) flag when independence failure produces unreliable predictions. When the independence assumption fails, Naive Bayes silently degrades. The model’s confidence scores become miscalibrated —it can be simultaneously confident and wrong.

SCX Diagnosis: The independence assumption $P(x_i|y) \perp P(x_j|y)$ resembles Assumption A2 (conditional expert independence), but applied to **features**, not experts. This is a category error. A2 requires independently-trained experts to have conditionally independent *errors* —a structural property of the audit system. Naive Bayes assumes the *data* satisfies A2, which is an empirical claim requiring verification that the model cannot perform.

Mathematical Audit: The true posterior under feature correlation is:

$$P(y|x) \propto P(y) \prod_i P(x_i|y) \cdot \prod_{i < j} \frac{P(x_i, x_j|y)}{P(x_i|y)P(x_j|y)}.$$

The correction factor $\prod_{i < j} P(x_i, x_j|y)/(P(x_i|y)P(x_j|y))$ is the product of pairwise dependence ratios. Naive Bayes sets this factor to 1 —an assumption with error magnitude $\varepsilon_{\text{indep}} = \|\text{true posterior} - \text{naive posterior}\|_{\text{TV}}$ that is never measured.

What SCX Provides: M independent Naive Bayes classifiers trained on disjoint feature subsets. YAJIE consensus across classifiers detects when feature correlation causes systematic disagreement. M_{eff} correction for residual feature-set overlap. Theorem 2 warning: the classifier cannot verify its own independence assumption. CERCIS Score explicitly penalizes unverified assumptions: $Q_{\text{NB}} = 0.35$, with the penalty proportional to estimated $\varepsilon_{\text{indep}}$.

4.4 4. Decision Trees Decision Trees

Verdict: **UNDECLARED** — $M = 1$, Overfits With No Audit

Why Popular: Interpretable. A decision tree is a flowchart of binary splits: if $x_j > \tau$, go to left child; else go to right child. Every split is explainable in natural language. CART (Breiman et al., 1984), ID3 (Quinlan, 1986), C4.5 (Quinlan, 1993) —decades of refinement. The tree structure is human-readable. No black box. Feature importance is directly visible from split frequencies.

Failure Mode: Overfitting with no audit mechanism. A single tree can grow until it perfectly memorizes the training data —one leaf per training point, zero training error, catastrophic test error. Pruning heuristics (minimum samples per leaf, maximum depth, cost-complexity pruning with parameter α) are **heuristics**, not theorems. There is no formal guarantee that the pruned tree generalizes. The tree’s confidence $p(y|x) = n_y/n_{\text{leaf}}$ is computed from training points that reached the same leaf —self-audit: the data that built the tree also certifies it (Theorem 2 violation).

SCX Diagnosis: $M = 1$. One tree, one opinion, one path from root to leaf. If that path is wrong, no dissenting voice exists. Theorem 1: single-expert detection bound applies. The tree’s Gini impurity or entropy reduction during splitting measures training data fit, not generalization quality. The tree has no mechanism to estimate its own test error without a held-out set —and even then, evaluating on held-out data is self-audit if the same architecture/hyperparameters are chosen based on that evaluation.

Mathematical Audit: A decision tree with L leaves partitions the input space into L regions $R_1, \dots, R_L$. The prediction in region R_ℓ is $\hat{y}_\ell = \frac{1}{|R_\ell|} \sum_{i \in R_\ell} y_i$. The variance of this estimate is $\sigma^2/|R_\ell|$. When $|R_\ell| = 1$, variance is unbounded $\rightarrow$ extreme overfitting. Cost-complexity pruning penalizes tree size: $\min_T \sum_\ell \sum_{i \in R_\ell} (y_i - \hat{y}_\ell)^2 + \alpha|T|$. This is a regularization heuristic —it reduces variance at the cost of bias, with α chosen by cross-validation (another heuristic).

What SCX Provides: Multi-tree consensus $\rightarrow$ Random Forest (Section 6.2). M independently-trained trees with bootstrap samples + random feature subsets satisfy A1 and A2. The YAJIE consensus across M trees provides Theorem 1 error detection. CERCIS Score of a single decision tree: $Q = 0$, $S = \eta N \leq 0.02$ —the tree itself provides zero quality guarantee.

4.5 5. SVM + Kernel Methods SVM + Kernel Methods

Verdict: **UNDECLARED** —Implicit Situs Without Lipschitz Guarantee

Why Popular: Max-margin theory. SVM finds the hyperplane that maximizes the margin between classes:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y_i(w^\top \phi(x_i) + b) \geq 1, \forall i.$$

The kernel trick $K(x, x') = \langle \phi(x), \phi(x') \rangle$ enables implicit infinite-dimensional feature maps. The representer theorem guarantees the solution $w = \sum_i \alpha_i y_i \phi(x_i)$ is a sparse combination of support vectors. Dual formulation: $\max_\alpha \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$ subject to $\sum_i \alpha_i y_i = 0, \alpha_i \geq 0$. Theoretically beautiful. Practically effective on small-to-medium datasets.

Failure Mode: **Kernel choice is arbitrary; encoding quality is unmeasured.**

The RBF kernel $K(x, x') = \exp(-\gamma \|x - x'\|^2)$ is the default. Why? Because it works empirically, not because it's certified. The bandwidth γ is tuned by cross-validation — the **same data** used for both training and evaluation. The implicit feature map ϕ has Lipschitz constant $\text{Lip}(\phi) \approx \sqrt{2\gamma}$ for the RBF kernel, which is controlled only by the heuristic choice of γ .

SCX Diagnosis: The kernel trick $\phi(x)$ is an implicit SITUS encoding: it maps data into a space where linear separation is possible. Theorem Situs-1 requires: (1) bounded Lipschitz constant $\text{Lip}(\phi) \leq L$, (2) verifiable reconstruction quality $\|x - \psi(\phi(x))\| \leq \varepsilon_\phi$, (3) explicit encoding dimension d_z . SVM kernels satisfy none:

- **Lipschitz:** $\text{Lip}(\phi_{\text{RBF}}) = \sqrt{2\gamma e^{-1}} \approx 0.86\sqrt{\gamma}$ — depends on heuristic γ .
- **Reconstruction:** No decoder ψ exists for RBF kernel — $\phi(x)$ lives in an infinite-dimensional RKHS. Reconstruction error ε_ϕ is undefined.
- **Dimension:** Implicit and potentially infinite. The effective dimension (number of support vectors) is data-dependent and uncontrolled.

Mathematical Audit: The kernel matrix $K_{ij} = K(x_i, x_j)$ must be positive semi-definite. Mercer's theorem guarantees this for valid kernels. But PSD-ness is a **minimum** requirement — it doesn't guarantee encoding quality. A PSD kernel can still have unbounded Lipschitz constant, making the encoding unverifiable. The SVM's margin $\gamma_{\text{margin}} = 1/\|w\|$ is maximized during training, but this is a training-data property, not a generalization guarantee.

What SCX Provides: SITUS encoding with: explicit dimensionality d_z , Lipschitz bound $\text{Lip}(\phi) \leq L_\phi$, reconstruction error ε_ϕ , and quality score $Q_{\text{SITUS}} = (1 - \varepsilon_\phi) \exp(-L_\phi \sigma_x^2)$. M independent SVMs with different kernels form a multi-expert system. YAJIE consensus detects when kernel choice matters. SPRING records support vectors permanently, enabling retrospective audit of decision boundary changes.

4.6 6. k-Means Clustering k-Means Clustering

Verdict: **UNDECLARED** — Arbitrary K, Self-Referential Quality

Why Popular: Simple clustering. Lloyd's algorithm alternates between:

- **Assignment:** $c_i = \arg \min_{k \in [K]} \|x_i - \mu_k\|^2$
- **Update:** $\mu_k = \frac{1}{|C_k|} \sum_{i: c_i = k} x_i$

Converges to a local optimum of the within-cluster sum of squares (WCSS): $\text{WCSS} = \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2$. $O(nKd)$ per iteration. The workhorse of unsupervised learning since 1957.

Failure Mode: **K is chosen by elbow method — a heuristic, not a theorem.** The number of clusters K is a fundamental structural parameter. Choosing it by plotting

WCSS vs. K and visually identifying the “elbow” is subjective, unreproducible, and provides zero quality guarantee. Worse: k-means always finds K clusters even when the data has no cluster structure. The algorithm cannot declare “these data are not clusterable” —it will impose K clusters on uniform noise.

SCX Diagnosis: State discovery without state quality guarantee. k-means partitions the state space into K Voronoi cells —a form of SITUS encoding. Theorem Situs-1 requires the encoding quality be measurable. But WCSS is self-referential: the algorithm optimizes what it measures. Theorem 2 violation: using WCSS to evaluate clustering quality is self-audit —k-means is judged by its own objective function.

Mathematical Audit: The quality of k-means clustering should be measured by:

- **Silhouette score:** $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ where $a(i)$ is mean intra-cluster distance and $b(i)$ is mean nearest-cluster distance. Range: $[-1, 1]$.
- **Davies-Bouldin index:** average similarity between each cluster and its most similar cluster.
- **Stability:** how much do cluster assignments change when the algorithm is re-run with different initialization?

None of these is a formal guarantee. They are descriptive statistics without confidence intervals.

Failure Mode 5: Adversarial Vulnerability — $M = 1$ Weakness. All single-model algorithms share an additional failure mode the ML literature calls “adversarial examples” —small, imperceptible perturbations δ with $\|\delta\| \leq \varepsilon$ that cause $\hat{y}(x + \delta) \neq \hat{y}(x)$. The existence of adversarial examples is a Theorem 1 consequence: with $M = 1$, there is no second opinion to flag that the prediction changed under perturbation. With $M > 1$ experts, an adversarial perturbation must simultaneously fool all M experts, and the probability of finding such a perturbation decays as $\exp(-2M\varepsilon^2)$. Adversarial robustness is an audit property, not a training property.

The Path to SCX Compliance for Classical Algorithms. Each classical algorithm can be retrofitted with SCX audit. The recipe: (1) train M independent copies on disjoint bootstrap samples, (2) implement YAJIE consensus with inverse-variance weights, (3) deploy SPRING memory for permanent prediction-outcome logging, (4) compute and publish the CERCIS Score. The mathematical machinery is Theorem 1 (Hoeffding bound on multi-expert miss probability). The only cost is $M \times$ training cost —a small price for verifiability.

5 Part II: Ensemble Era (2000s) —The Unconscious Theorem 1

6 Part II: The Ensemble Era — Unconscious Theorem 1

The ensemble era discovered empirically what SCX proves formally: multiple models are better than one. Bagging, boosting, and random forests all use multiple “experts” —but they differ critically in whether those experts are *independent*. This distinction separates the verified from the guilty.

6.1 7. Bagging (Bootstrap Aggregating) Bagging

Verdict: **PARTIAL** —Implicit A1, No Formal M_{eff}

Why Popular: Reduces variance. Breiman (1996): train B models on B bootstrap samples, average predictions. For squared error loss, if base models have variance σ^2 and pairwise correlation ρ , the ensemble variance is:

$$\text{Var}(\hat{y}_{\text{bag}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

When $\rho = 0$, variance reduces by factor $1/B$. Bootstrap samples are drawn with replacement —each contains approximately 63.2% of the original data, with the remaining 36.8% serving as out-of-bag validation.

Failure Mode: **No formal M_{eff} guarantee.** Bootstrap samples overlap —the 63.2% overlap creates correlation between base models. The inter-model correlation $\bar{\rho}$ depends on model complexity, data distribution, and sample size. Bagging does not measure $\bar{\rho}$, does not declare M_{eff} , and provides no bound on the residual variance. The user gets a point estimate with no quality certificate.

SCX Diagnosis: Bagging implicitly implements A1 through bootstrap sampling, but:

1. The “independence” is approximate —bootstrap samples share data, so $I(\mathcal{D}_m; \mathcal{D}_{m'}) > 0$.
2. $M_{\text{eff}} = B/(1 + \bar{\rho})$ is undeclared —the user doesn’t know the effective number of independent experts.
3. No Hoeffding bound is provided —the user doesn’t know $\mathbb{P}(\text{miss})$.

Mathematical Audit: For $B = 100$ bootstrap models with inter-model error correlation $\bar{\rho} = 0.15$, the effective multiplicity is $M_{\text{eff}} = 100/1.15 \approx 87$. The error detection probability is:

$$\mathbb{P}(\text{detect}) \geq 1 - \exp(-2 \cdot 87 \cdot \Delta^2).$$

For $\Delta = 0.5$: $\mathbb{P}(\text{detect}) \geq 1 - \exp(-43.5) \approx 1 - 1.3 \times 10^{-19}$ —effectively certain. But this bound **only holds if Bagging measures and declares $\bar{\rho}$** , which it doesn't.

What SCX Provides: Explicit M declaration with measured $\bar{\rho}$. Formal Hoeffding bound. SPRING monitoring whether $\bar{\rho}$ drifts over time. YAJIE weighted voting instead of simple averaging. CERCIS Score: $Q_{\text{Bagging}} = 0.6$ (structural $M > 1$ but undeclared).

6.2 8. Random Forest Random Forest

Verdict: **VERIFIED** —Best Classical Algorithm, Unconsciously Implements Theorem 1

Why Popular: Robust, accurate, virtually hyperparameter-free. Breiman (2001) combined two randomization mechanisms:

1. **Bootstrap samples:** Each tree trained on a different bootstrap sample (A1: independent data subsets).
2. **Random feature subsets:** At each split, only $m = \lfloor \sqrt{p} \rfloor$ randomly chosen features are considered. This reduces inter-tree correlation while maintaining individual tree accuracy.

OOB (out-of-bag) error provides built-in validation: each tree is evaluated on the $\approx 36.8\%$ of data not in its bootstrap sample. No need for a separate validation set.

Failure Mode: Random Forest's failures are sins of omission, not commission:

1. It does not declare M —though the number of trees is explicit, it's not connected to an error detection guarantee.
2. OOB error is a point estimate without confidence intervals. The user doesn't know $\mathbb{P}(\text{OOB error} > \text{true error} + \delta)$.
3. No online drift detection —when data distribution shifts, the forest degrades silently.
4. No CERCIS Score —users cannot compare forest quality across different configurations or datasets.

SCX Diagnosis: Random Forest **accidentally implements Theorem 1**. The structure:

- B trees trained on bootstrap samples $\rightarrow$ A1 (independent training data) partially satisfied.
- Random feature subsets at each split $\rightarrow$ A2 (conditional independence through diverse perspectives) partially satisfied.
- OOB error = cross-audit $\rightarrow$ each tree is evaluated by data it never saw, providing independent quality estimates.

The probability that all B trees miss an error decays as:

$$\mathbb{P}(\text{all miss}) \leq \exp(-2M_{\text{eff}}\Delta^2), \quad M_{\text{eff}} = \frac{B}{1 + \bar{\rho}},$$

where $\bar{\rho}$ is the average inter-tree error correlation (typically 0.05–0.2 for Random Forest due to double randomization). For $B = 500$, $\bar{\rho} = 0.1$, $M_{\text{eff}} = 500/1.1 \approx 455$.

Mathematical Audit: The CERCIS Score for Random Forest:

$$Q_{\text{RF}} = \frac{1}{5} [q_1 + q_2 + q_3 + q_{\text{SITUS}} + q_{\text{SPRING}}] \quad (11)$$

$$= \frac{1}{5} [1.0 \text{ (Thm 1: } M > 1) + 0.8 \text{ (Thm 2: OOB avoids self-audit)} \quad (12)$$

$$+ 0.5 \text{ (Thm 3: trees are shallow, limited depth issue)} \quad (13)$$

$$+ 0.7 \text{ (Situs-1: tree splits = state partition, implicit encoding)} \quad (14)$$

$$+ 0.0 \text{ (Spring-1: no permanent memory)}] \quad (15)$$

$$= 0.60. \quad (16)$$

With $N = 0.2$ (widely used, well-validated): $\text{CERCIS} = 0.60 + 0.2 \cdot 0.2 = 0.64$. But this is *conservative*. The full SCX-augmented Random Forest:

$$Q_{\text{SCX-RF}} = \frac{1}{5} [1.0 + 1.0 + 0.8 + 0.7 + 0.5] = 0.80, \quad (17)$$

$$\text{CERCIS}_{\text{SCX-RF}} = 0.80 + 0.2 \cdot 0.2 = 0.84. \quad (18)$$

What SCX Provides: Explicit CERCIS Score: $Q_{\text{RF}} = 1 - \exp(-2M_{\text{eff}}\Delta^2)$ combined with OOB accuracy. SPRING for online drift detection: permanently records per-tree prediction distributions, flags when inter-tree disagreement patterns change. YAJIE Nash equilibrium audit with tree weights proportional to OOB performance. Cryptographic hash binding of forest structure and training data manifest.

6.3 9. Boosting Boosting — AdaBoost/GBDT/XGBoost/LightGBM

Verdict: **UNDECLARED** — Sequential Dependency Violates A1

Why Popular: State-of-the-art on tabular data. The boosting family iteratively trains weak learners, each focusing on predecessors' errors:

- **AdaBoost** (Freund & Schapire, 1995): Re-weights misclassified examples exponentially.
- **Gradient Boosting** (Friedman, 2001): Each new tree fits the negative gradient of the loss: $h_t = \arg \min_h \sum_i (-\partial(y_i, F_{t-1}(x_i))/\partial F_{t-1}(x_i) - h(x_i))^2$.
- **XGBoost** (Chen & Guestrin, 2016): Adds regularization $\Omega(h) = \gamma T + \frac{1}{2}\lambda \|w\|^2$, second-order approximation, system optimization.
- **LightGBM** (Ke et al., 2017): Gradient-based One-Side Sampling (GOSS), Exclusive Feature Bundling (EFB).

These dominate Kaggle and industrial tabular ML.

Failure Mode: Each new expert depends on previous errors $\rightarrow$ VIOLATES A1. This is the fundamental structural flaw. Expert f_t is trained to correct the residuals of $f_1, \dots, f_{t-1}$:

$$f_t = \arg \min_f \sum_i (y_i, F_{t-1}(x_i) + f(x_i)).$$

The training objective for expert t is explicitly a function of experts $1, \dots, t-1$. This sequential dependency breaks A1 (independent training) and A2 (conditional independence):

$$\mathbb{P}(\hat{y}_t \mid y, \hat{y}_1, \dots, \hat{y}_{t-1}) \neq \mathbb{P}(\hat{y}_t \mid y).$$

SCX Diagnosis: Sequential dependency means the effective multiplicity M_{eff} is **much smaller** than the number of weak learners T . In the limit, $T \rightarrow \infty$ with strong dependency, $M_{\text{eff}} \rightarrow 1$. This is why boosting overfits where Random Forest doesn't: sequential experts learn to model noise **together**, producing correlated errors that no consensus mechanism can detect. Each tree corrects the previous trees' noise, creating a chain of noise compensation that appears as "learning" but is actually memorization.

Mathematical Audit: Let $\rho_{t,t'} = (\mathbf{1}[f_t \text{ errs}], \mathbf{1}[f_{t'} \text{ errs}])$. For sequential training, $\rho_{t,t'} \geq \rho_0 > 0$ for $t \neq t'$ due to shared dependency on earlier errors. The effective multiplicity:

$$M_{\text{eff}} = \frac{T}{1 + \frac{2}{T(T-1)} \sum_{t < t'} \rho_{t,t'}} \leq \frac{T}{1 + \rho_0}.$$

For typical boosting with $\rho_0 \approx 0.4$: $M_{\text{eff}} \leq T/1.4$. With $T = 100$: $M_{\text{eff}} \leq 71$. Much worse: the sequential structure means $\rho_{t,t'}$ increases with $|t - t'|$, so later trees are *more* correlated, not less. The effective multiplicity may be as low as $M_{\text{eff}} \approx 10-20$ for $T = 100$.

What SCX Provides: PARALLEL experts, not sequential. Train M independent boosting chains on disjoint data, then apply YAJIE consensus **across** chains. Each chain may internally violate A1 (sequential within-chain), but chains are independent $\rightarrow$ A1 satisfied at meta-level. CERCIS Score explicitly penalizes sequential dependency: $Q_{\text{boosting}} = 0.3$ vs. $Q_{\text{RF}} = 0.6$. The Q gap explains why boosting empirically overfits more: lower effective multiplicity.

6.4 10. Stacking (Stacked Generalization) Stacking

Verdict: **PARTIAL** —Meta-Learner Has No Audit

Why Popular: Learns to combine experts optimally. Wolpert (1992): Train M base learners $f_1, \dots, f_M$ on training data. Use their predictions $\hat{y}_i^{(m)} = f_m(x_i)$ as features for a meta-learner g trained on validation data:

$$\hat{y}_{\text{stack}} = g(f_1(x), f_2(x), \dots, f_M(x)).$$

Instead of fixed averaging or voting, stacking learns the optimal combination function. Can use any model as meta-learner: linear regression, neural network, gradient boosting. **Failure Mode:** The meta-learner introduces an unverifiable component. The meta-learner g is a single model making the final decision — $M = 1$ at the critical decision point. If g overfits the base-learner outputs, there is no detection mechanism. The base learners provide diversity, but the meta-learner collapses that diversity into a single output. Worse: if g learns to trust a consistently wrong base learner, the consensus actually *amplifies* error.

SCX Diagnosis: The meta-learner resembles YAJIE with learned weights. But YAJIE requires: (1) weights derived from verifiable competence scores ω_m , (2) the weight mechanism itself is a fixed mathematical function (Nash equilibrium), (3) the weight-learning process has its own M declaration. Stacking’s meta-learner satisfies none.

Mathematical Audit: For M base learners with errors $\varepsilon_m = \mathbb{P}(f_m(x) \neq y)$, the optimal combination weight (inverse-variance) is $w_m \propto 1/\text{Var}(\varepsilon_m)$. Stacking estimates these weights from data. The estimation error $\|\hat{w} - w^*\|$ is unknown and unverifiable. If the meta-learner overfits, $\|\hat{w} - w^*\| \gg 0$, and the stacked prediction is worse than simple averaging.

What SCX Provides: YAJIE Nash equilibrium audit: each base learner has verifiable competence ω_m on held-out data. YAJIE computes Pareto-optimal weights that no coalition can improve by deviation. The mechanism is a fixed function —no learned parameters, no meta-overfitting. SPRING permanently records base-learner and meta-learner performance. M_{meta} declaration for the meta-learner’s own audit.

7 Part III: Deep Learning Revolution (2012–2017) —Depth as Compensation

8 Part III: The Deep Learning Revolution — Depth as Compensation

Deep learning succeeded by stacking many layers. The standard narrative: depth enables hierarchical feature learning —early layers learn edges, middle layers learn parts, late layers learn objects. This narrative is comforting but unverified. SCX Theorem 3 reveals a darker possibility: depth may be compensating for unmodeled noise, not learning genuine structure. The architectural innovations of 2012–2017 can be understood as *accidental* implementations of SCX principles, with varying degrees of success.

8.1 11. Deep MLPs Deep MLPs

Verdict: **UNDECLARED** —Depth Without Purpose

Why Popular: Universal approximation. The Universal Approximation Theorem proves that a single hidden layer with sufficiently many neurons can approximate any continuous function on a compact set. Deep MLPs stack multiple hidden layers:

$$h_0 = x, \quad h_\ell = \sigma(W_\ell h_{\ell-1} + b_\ell), \quad \ell = 1, \dots, L,$$

with nonlinear activation σ (ReLU, tanh, sigmoid). Backpropagation efficiently computes gradients via the chain rule. Theoretically capable of learning arbitrary functions with sufficient capacity.

Failure Mode: **Vanishing gradients + no depth justification.** As L increases, gradients decay exponentially:

$$\frac{\partial}{\partial W_1} = \frac{\partial}{\partial h_L} \cdot \prod_{\ell=2}^L \frac{\partial h_\ell}{\partial h_{\ell-1}} \cdot \frac{\partial h_1}{\partial W_1}.$$

For sigmoid: $\partial h_\ell / \partial h_{\ell-1} = \sigma'(z_\ell) W_\ell \leq 0.25 \cdot \|W_\ell\|$. The product of L terms each ≤ 0.25 vanishes as $L \rightarrow \infty$. ReLU mitigates ($\sigma'(z) = 1$ for $z > 0$) but introduces dead neurons.

SCX Diagnosis: Theorem 3: stacking L layers without skip connections means each layer transforms the previous layer’s output. $h_\ell = T_\ell(h_{\ell-1})$. If layer ℓ corrects an error from layer $\ell - 1$, that error might be genuine modeling error or training noise. Without independent audit of each layer’s contribution, depth is epistemically opaque.

Mathematical Audit: The effective depth L_{eff} —the number of layers that *genuinely* improve generalization —is unknown. A 100-layer MLP might have $L_{\text{eff}} \approx 10$ with 90 layers compensating for noise. The unidentifiability is formal:

$$\nexists \text{ test } T : \mathcal{H} \rightarrow \mathbb{N} \text{ such that } T(f) = L_{\text{eff}}(f)$$

without access to $M > 1$ independently-trained copies. The model’s own training/validation loss is self-audit (Theorem 2).

What SCX Provides: Explicit noise audit: for each layer ℓ , compute the SITUS reconstruction quality of $h_{\ell-1} \rightarrow h_\ell$. If h_ℓ can be reconstructed from $h_{\ell-1}$ with error $< \tau$, layer ℓ is redundant. Required depth: $L_{\min} = \min\{\ell : \|h_\ell - h_{\ell-1}\|_{\text{recon}} > \tau\}$. CERCIS Score penalizes unjustified depth: Q decreases as $(L - L_{\min})/L$.

8.2 12. CNN (Convolutional Neural Networks) CNN

Verdict: **PARTIAL** —Built-in Situs, No Lipschitz Guarantee

Why Popular: Revolutionized computer vision. The convolution operation:

$$(F * K)[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k F[i + u, j + v] \cdot K[u, v],$$

exploits translation equivariance: shifting the input shifts feature maps identically. Shared weights across spatial positions $\rightarrow O(k^2 c_{\text{in}} c_{\text{out}})$ parameters instead of $O(H^2 W^2 c_{\text{in}} c_{\text{out}})$. AlexNet (2012) won ImageNet by 10% margin. VGG (2014), Inception (2014–2016), ResNet (2015) —continuous improvement.

Failure Mode: No Lipschitz guarantee on the equivariance. Translation equivariance $f(T_v x) = T_v f(x)$ holds exactly only for continuous convolution with infinite support. Discrete convolution with stride > 1 , padding, and boundary effects breaks exact equivariance. More importantly: $\text{Lip}(f_{\text{conv}}) = \|W\|_{\text{spectral}}$ —the spectral norm of the convolution kernel. During training, $\|W\|_{\text{spectral}}$ can grow unboundedly. A small adversarial perturbation δ with $\|\delta\| \leq \varepsilon$ can produce $\|f(x + \delta) - f(x)\| \geq \|W\|_{\text{spectral}} \cdot \varepsilon$, which can be arbitrarily large.

SCX Diagnosis: Convolution = SITUS spatial encoding. Translation equivariance = encoding preserves spatial relationships. Theorem Situs-1 requires: (1) bounded Lipschitz $\text{Lip}(\phi) \leq L$, (2) verifiable reconstruction. Standard CNNs have neither. Max-pooling: $\text{MaxPool}(F)[i, j] = \max_{u,v \in \{0,1\}} F[2i + u, 2j + v]$ —this destroys information, making reconstruction impossible. The encoding quality Q_{SITUS} cannot be computed.

Mathematical Audit: The Lipschitz constant of a CNN layer is:

$$\text{Lip}(f_\ell) = \|W_\ell\|_{\text{spectral}} = \sup_{x \neq 0} \frac{\|W_\ell * x\|_2}{\|x\|_2} = \max_k |\hat{W}_\ell[k]|,$$

where $\hat{W}_\ell$ is the Fourier transform of the kernel. For a 3×3 kernel with weights in $[-1, 1]$, $\text{Lip} \leq 9$ (worst case). For a 7×7 kernel, $\text{Lip} \leq 49$. Deep CNNs with L layers have $\text{Lip}(f) \leq \prod_{\ell=1}^L \text{Lip}(f_\ell)$, which can be astronomically large. Spectral normalization (Miyato et al., 2018) constrains $\text{Lip}(f_\ell) \leq 1$ but is rarely used in practice.

What SCX Provides: SITUS Lipschitz bound via spectral normalization on every convolutional layer. Invertible SITUS encoding: replace max-pooling with invertible downsampling (e.g., pixel shuffle, wavelet transform). Encoding quality score $Q_{\text{SITUS}} = (1 - \varepsilon_\phi) \exp(-L_\phi \sigma_x^2)$. YAJIE consensus across M CNNs with different architectures verifies robustness of spatial encoding.

8.3 13. RNN / LSTM RNN / LSTM

Verdict: **UNDECLARED** —Memory Decay = Bug, Not Feature

Why Popular: Sequence modeling. RNN: $h_t = \sigma(W_h h_{t-1} + W_x x_t + b)$. LSTM (Hochre-

iter & Schmidhuber, 1997) added gating:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (\text{forget gate}) \quad (19)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (\text{input gate}) \quad (20)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (\text{candidate}) \quad (21)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{cell state}) \quad (22)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (\text{output gate}) \quad (23)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{hidden state}) \quad (24)$$

GRU (Cho et al., 2014) simplified to two gates. LSTMs dominated NLP for years before Transformers.

Failure Mode: LSTM’s forget gate = Spring memory WITH voluntary forgetting. But Spring NEVER forgets. $f_t \in (0, 1)$ scales down the previous cell state. When $f_t \approx 0$, stored information is **permanently erased**. There is no recovery mechanism, no audit trail, no record that information was discarded. This is presented as a feature —the network learns what is and isn’t important. From an audit perspective: **catastrophic**. Evidence is destroyed.

SCX Diagnosis: Theorem Spring-1: $M_t \supseteq M_{t-1}$ —memory must be monotonic. LSTM’s cell state update $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$ allows $f_t < 1$, meaning $\|c_t\|_2$ can decrease —memory SHRINKS. Furthermore, LSTM has bounded memory: $c_t \in \mathbb{R}^{d_c}$ is a fixed-size vector. When new information arrives, old information must be overwritten (even if $f_t \approx 1$, the additive term $i_t \odot \tilde{c}_t$ dilutes old information). This is bounded-capacity memory with overwriting —the opposite of SPRING’s unbounded, append-only memory.

Mathematical Audit: The memory capacity of LSTM at time t is:

$$\text{Capacity}(c_t) = \|c_t\|_2 \leq \|c_{t-1}\|_2 + \|\tilde{c}_t\|_2 \leq \sum_{\tau=1}^t \|\tilde{c}_\tau\|_2.$$

This is bounded by $d_c \cdot \max_\tau \|\tilde{c}_\tau\|_\infty$, which is finite and independent of t . The effective memory horizon —how far back information is retained —is finite and data-dependent. Information from time $t - \Delta t$ is retained with probability $\prod_{\tau=t-\Delta t}^t f_\tau$, which decays exponentially with Δt .

What SCX Provides: SPRING permanent memory: $M_t = M_{t-1} \cup \{(x_t, y_t, \hat{y}_t, \text{audit}_t)\}$. Every observation stored permanently. No forget gate. No capacity limit. No information loss. SPRING provides an external memory that the prediction model queries (like Transformer’s cross-attention but with unbounded memory). YAJIE consensus operates on SPRING’s complete history, not a lossy hidden state. CERCIS Score: $Q_{\text{LSTM}} = 0.15$ —penalized for memory decay.

8.4 14. ResNet (Residual Networks) ResNet

Verdict: **VERIFIED** —Explicit Signal-Noise Separation

Why Popular: Enabled 152-layer networks. He et al. (2015) proposed residual connections:

$$h_{\ell+1} = h_{\ell} + F_{\ell}(h_{\ell}; W_{\ell}).$$

The identity skip connection preserves the signal h_{ℓ} ; the residual block F_{ℓ} learns the *correction*. Gradient flow: $\partial h_{\ell+1}/\partial h_{\ell} = I + \partial F_{\ell}/\partial h_{\ell}$ —the identity term prevents vanishing gradients. Won ImageNet 2015. Became the default architecture for vision.

Failure Mode: ResNet gets the structure right but not the audit. $h_{\ell+1} = h_{\ell} + F_{\ell}(h_{\ell})$ separates signal (h_{ℓ}) from correction (F_{ℓ}). This is exactly what Theorem 3 demands: a clean signal baseline + independently-auditable correction path. But ResNet doesn't audit the corrections. It doesn't verify that F_{ℓ} genuinely corrects error vs. injecting noise. It doesn't declare L_{eff} or provide a depth formula.

SCX Diagnosis: **This is the ONLY architecture that formally separates signal from noise.** The skip connection h_{ℓ} carries the clean signal baseline; $F_{\ell}(h_{\ell})$ is the noise correction path. Theorem 3 says: with one model, you cannot tell if F_{ℓ} corrects error or noise. But ResNet's structure *enables* the audit: you can measure $\|F_{\ell}(h_{\ell})\|_2$ —the magnitude of the correction. If $\|F_{\ell}(h_{\ell})\|_2 \approx 0$ for most inputs, the layer is unnecessary. If $\|F_{\ell}(h_{\ell})\|_2$ varies wildly across inputs, the correction is input-dependent (potentially noise compensation).

Mathematical Audit: The residual block's contribution can be audited:

$$\text{CorrectionRatio}_{\ell} = \frac{\mathbb{E}_x[\|F_{\ell}(h_{\ell})\|_2]}{\mathbb{E}_x[\|h_{\ell}\|_2]}.$$

If $\text{CorrectionRatio}_{\ell} < \tau$, layer ℓ contributes negligible correction $\rightarrow$ can be pruned. If $\text{CorrectionRatio}_{\ell} \gg 1$, the residual dominates the skip connection $\rightarrow$ the original signal is being overwritten, not refined. The SCX-augmented ResNet depth formula:

$$L_{\min} = \min \left\{ L : \sum_{\ell=1}^L \text{CorrectionRatio}_{\ell} \geq \frac{1}{\eta} \right\},$$

where η is the target residual error.

What SCX Provides: Explicit depth formula $L_{\min} = O(\log(1/\eta))$. Per-block audit: M experts train the same block with different initializations; YAJIE consensus verifies the correction is reproducible. SPRING records which blocks contributed signal vs. noise. CERCIS Score: $Q_{\text{ResNet}} = 0.85$ —highest of any deep architecture.

8.5 15. Batch Normalization Batch Normalization

Verdict: **VERIFIED** —Satisfies A5 (Bounded State Distribution)

Why Popular: Stabilizes training. Ioffe & Szegedy (2015): for each mini-batch :

$$\mu = \frac{1}{\parallel} \sum_{i \in} x_i, \quad \sigma^2 = \frac{1}{\parallel} \sum_{i \in} (x_i - \mu)^2,$$

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}, \quad y_i = \gamma \hat{x}_i + \beta.$$

Reduces internal covariate shift. Enables higher learning rates (10×+). Acts as mild regularizer. Standard in virtually all architectures.

Failure Mode: **Breaks at small batch sizes.** When $\parallel$ is small (e.g., 2 for large models), μ and σ^2 are noisy estimators. The variance of σ^2 is approximately $2\sigma^4/(\parallel - 1)$, which diverges as $\parallel \rightarrow 1$. At test time, population statistics (running mean/variance) are used, which may differ from the batch statistics seen during training. This is a distribution mismatch that BatchNorm doesn't detect —it just uses the stale running statistics.

SCX Diagnosis: BatchNorm enforces $\mathbb{E}[h_\ell] \approx 0, \text{Var}[h_\ell] \approx 1 \rightarrow$ state distribution is bounded $\rightarrow$ A5 is structurally satisfied. The Lipschitz constant of the normalized layer is $\text{Lip}(f_{\text{BN}}) = \|\gamma/\sigma\|_\infty$, which is controlled by the learned scale γ . But BatchNorm doesn't **monitor** the distribution —it normalizes without recording whether normalization is still valid.

Mathematical Audit: The distribution shift between training and test can be measured:

$$\text{KL}(\mathcal{N}(\mu_{\text{train}}, \sigma_{\text{train}}^2) \parallel \mathcal{N}(\mu_{\text{test}}, \sigma_{\text{test}}^2)) = \frac{1}{2} \left[\frac{\sigma_{\text{train}}^2}{\sigma_{\text{test}}^2} + \frac{(\mu_{\text{test}} - \mu_{\text{train}})^2}{\sigma_{\text{test}}^2} - 1 + \ln \frac{\sigma_{\text{test}}^2}{\sigma_{\text{train}}^2} \right].$$

BatchNorm doesn't compute this. It just applies the stale normalization.

What SCX Provides: SPRING state distribution drift monitor. Permanently records (μ_t, σ_t^2) at each layer for each batch. When $\text{KL}(\rho_t \parallel \rho_{\text{train}}) > \tau$, SPRING triggers distribution shift alert. CERCIS Score incorporates distribution stability penalty. YAJIE consensus across M independently-normalized copies detects when normalization fails.

8.6 16. Dropout

Verdict: **VERIFIED** — 2^n Implicit Sub-Networks Voting

Why Popular: Simple, effective regularization. Srivastava et al. (2014): during

training, randomly drop each neuron with probability p :

$$h_\ell = \text{Bernoulli}(1 - p) \odot \sigma(W_\ell h_{\ell-1} + b_\ell).$$

At test time, use all neurons with weights scaled by $(1 - p)$:

$$h_\ell^{\text{test}} = (1 - p) \cdot \sigma(W_\ell h_{\ell-1} + b_\ell).$$

This approximates averaging an ensemble of 2^n sub-networks (where n is the number of dropout neurons). Prevents co-adaptation.

Failure Mode: Dropout accidentally implements Theorem 1 but doesn’t declare it. The test-time scaling $w_{\text{test}} = (1 - p)w_{\text{train}}$ approximates the geometric mean of all 2^n sub-network predictions under a Gaussian approximation. But: (1) the sub-networks are **highly correlated** —they share all weights and differ only in which neurons are dropped, (2) $M_{\text{eff}} \ll 2^n$, (3) no formal Hoeffding bound is provided.

SCX Diagnosis: Theorem 1 directly applies: each sub-network is an “expert” voting at test time. The effective multiplicity depends on the dropout rate and network structure:

$$M_{\text{eff}} = \frac{2^n}{1 + \bar{\rho}_{\text{dropout}}},$$

where $\bar{\rho}_{\text{dropout}}$ is the average output correlation between sub-networks. Since sub-networks share weights, $\bar{\rho}_{\text{dropout}}$ is high. For a network with $n = 1000$ dropout neurons, $p = 0.5$, and $\bar{\rho}_{\text{dropout}} = 0.99$: $M_{\text{eff}} = 2^{1000}/1.99 \approx \infty$ —but only if independence holds, which it emphatically doesn’t. A more realistic estimate: $M_{\text{eff}} \approx 10 - 100$ due to weight sharing.

Mathematical Audit: MC Dropout (Gal & Ghahramani, 2016) keeps dropout on at test time with K stochastic forward passes. This provides K predictions $\hat{y}^{(1)}, \dots, \hat{y}^{(K)}$ that estimate predictive uncertainty. The variance $\text{Var}[\hat{y}^{(k)}]$ is an uncertainty estimate — but it’s self-audited (Theorem 2): the model estimates its own uncertainty using its own stochasticity. The uncertainty decomposition:

$$\text{Var}[\hat{y}] = \underbrace{\text{Var}_{\text{model}}[\mathbb{E}[\hat{y}|x, \theta]]}_{\text{epistemic}} + \underbrace{\mathbb{E}_{\text{model}}[\text{Var}[\hat{y}|x, \theta]]}_{\text{aleatoric}}$$

is valid only if the dropout distribution approximates the true posterior $p(\theta|\mathcal{D})$, which is unverifiable.

What SCX Provides: Explicit M_{eff} measurement: sample K sub-networks, compute pairwise output correlation $\bar{\rho}$, declare $M_{\text{eff}} = K/(1 + \bar{\rho})$. YAJIE consensus with sub-network votes weighted by validation performance. CERCIS Score: $Q_{\text{Dropout}} = 0.8$ —high because structurally provides $M > 1$.

8.7 17. Attention / Transformer Attention / Transformer

Verdict: **VERIFIED** —Multi-Head = Spring Gatekeepers, Bounded Memory is the Flaw

Why Popular: Dominates NLP and CV. Vaswani et al. (2017):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

Multi-head attention runs h parallel operations:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad \text{MultiHead} = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O.$$

The Transformer replaced recurrence with self-attention. BERT (2018), GPT-3 (2020), ViT (2020), GPT-4 (2023) —the foundation of modern AI.

Failure Mode: Transformer is **structurally brilliant** from an SCX perspective but has one critical flaw: the KV cache is bounded. Self-attention computes attention over a context window of size L (e.g., 2048 for GPT-3, 128K for GPT-4). Keys and values from before the context window are **discarded**. This is SPRING with voluntary forgetting — Theorem Spring-1 violation.

SCX Diagnosis: $QK^\top$ = content-addressable memory query $\rightarrow$ SPRING. Multi-head = $M = h$ parallel SPRING gatekeepers operating simultaneously. Softmax = probabilistic gating over stored keys. The attention pattern $\alpha_{ij} = \text{softmax}(q_i^\top k_j / \sqrt{d_k})$ is a probability distribution over memory locations —SPRING’s gating mechanism. Theorem Spring-1 is **structurally satisfied** by multi-head attention but **capacity-violated** by the bounded context window.

Mathematical Audit: The effective multiplicity of multi-head attention:

$$M_{\text{eff}} = \frac{h}{1 + \bar{\rho}_{\text{head}}},$$

where $\bar{\rho}_{\text{head}}$ is the average attention pattern correlation between heads. Michel et al. (2019) showed that many attention heads can be pruned without performance loss, suggesting $\bar{\rho}_{\text{head}}$ is high and $M_{\text{eff}} \ll h$. For $h = 16$ and $\bar{\rho}_{\text{head}} = 0.7$: $M_{\text{eff}} = 16/1.7 \approx 9.4$.

What SCX Provides: SPRING’s permanent memory growth: maintain an unbounded SPRING memory bank. Keys and values from all past sequences retained permanently. New sequences query the entire SPRING memory. Computational cost addressed via efficient approximate nearest neighbor search. YAJIE consensus across heads with measured correlation. CERCIS Score: $Q_{\text{Transformer}} = 0.8$ —penalized by 0.2 for bounded memory.

8.8 18. Transformer Variants: BERT, GPT, ViT Transformer Variants: BERT, GPT, ViT

BERT (2018) —Masked Language Modeling: **UNDECLARED** of Self-Audit

BERT pre-trains by masking 15% of tokens and predicting them from context. This is self-supervised learning: labels are generated from the data itself. Theorem 2: self-generated labels $\rightarrow$ self-audit $\rightarrow$ no quality guarantee on the learned representations. The MLM accuracy tells you how well BERT predicts held-out tokens, not whether those predictions encode genuine linguistic knowledge.

SCX Augmentation: M independently-masked versions of the same text, each with different random masks. YAJIE consensus across masked versions provides Theorem 1 detection of spurious predictions.

GPT (2018–2024) —Autoregressive Generation: **UNDECLARED** of $M = 1$

GPT generates tokens one at a time: $p(x_t|x_{<t})$. At each step, exactly one continuation is chosen. No second opinion. No verification of generation quality. Hallucinations are undetectable by the model itself. Theorem 1 with $M = 1$: no detection guarantee for generated falsehoods.

SCX Augmentation: M independent generation heads producing M candidate continuations. YAJIE consensus selects the most verified continuation. Factual claims cross-referenced against SPRING’s permanent knowledge base.

9 Part IV: Generative Models —The Audit Crisis

10 Part IV: Generative Models — The Audit Crisis

Generative models create new data: images, text, audio, video. They represent the frontier of ML capability —and the nadir of audit quality. The central problem: how do you verify the quality of something that never existed before? The answer, for current generative models: you don’t. And you can’t, because $M = 1$ or self-audit.

10.1 19. GAN (Generative Adversarial Networks) GAN

Verdict: **UNDECLARED** — $M = 1$ Discriminator Explains ALL GAN Failures

Why Popular: First realistic image generation. Goodfellow et al. (2014):

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))].$$

Generator G maps noise z to synthetic samples; Discriminator D distinguishes real from fake. At the Nash equilibrium of this minimax game, $G(z) \sim p_{\text{data}}$ —the generator perfectly mimics the data distribution.

Failure Mode: **ALL GAN failures are consequences of $M = 1$ discriminator.**

Theorem 1: $M_D = 1 \implies$ no detection guarantee. The discriminator is a single neural network. If G finds a pattern that fools this one network, there is no second opinion.

SCX Diagnosis: **Mode collapse** is the most famous GAN failure: the generator produces one type of output (e.g., one digit, one face pose) that reliably fools the discriminator. Why? Because one discriminator has one “perspective.” All the generator needs is one loophole. With $M > 1$ discriminators, each with different architecture/initialization, the generator must fool ALL of them simultaneously. The probability of finding one pattern that simultaneously fools M independent discriminators decays as $\exp(-2M\Delta^2)$.

Mathematical Audit: The GAN objective is a two-player zero-sum game. The value function:

$$V(G, D) = \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))].$$

For fixed G , the optimal discriminator is $D^*(x) = p_{\text{data}}(x)/(p_{\text{data}}(x) + p_G(x))$. The generator’s objective becomes $\min_G \text{JSD}(p_{\text{data}} \| p_G) - \log 4$, where JSD is Jensen-Shannon divergence. In theory, G minimizes JSD. In practice: D is a neural network with finite capacity $\rightarrow D$ cannot achieve $D^* \rightarrow$ the generator optimizes against an imperfect discriminator $\rightarrow$ the generator exploits imperfections.

With M discriminators $D_1, \dots, D_M$ trained independently:

$$V_M(G, D_1, \dots, D_M) = \frac{1}{M} \sum_{m=1}^M \left[\mathbb{E}_{x \sim p_{\text{data}}} [\log D_m(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D_m(G(z)))] \right].$$

The generator must now fool the **consensus** of discriminators. The probability of finding a sample that fools all M discriminators is:

$$\mathbb{P}(\text{fool all}) \leq \prod_{m=1}^M \mathbb{P}(\text{fool } D_m) \leq (\max_m \mathbb{P}(\text{fool } D_m))^M.$$

If each discriminator has 90% detection rate, $\mathbb{P}(\text{fool all}) \leq 0.1^M$. For $M = 5$: 10^{-5} . For $M = 1$: 0.1.

What SCX Provides: $M > 1$ independent discriminators. YAJIE consensus: generator must satisfy all discriminators simultaneously. SPRING permanently records generated samples and discriminator verdicts. CERCIS Score: $Q_{\text{GAN}} = 0.1$ ($M=1$). $Q_{\text{SCX-GAN}} =$

$1 - \exp(-2M\Delta^2)$ with $M > 1$.

10.2 20. VAE (Variational Autoencoders) VAE

Verdict: **UNDECLARED** —Self-Audit (Theorem 2)

Why Popular: Principled probabilistic framework. Kingma & Welling (2014): maximize ELBO

$$= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \| p(z)).$$

Encoder $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x))$ maps x to latent distribution; Decoder $p_\theta(x|z)$ reconstructs x from z . The KL term $\text{KL}(\mathcal{N}(\mu, \sigma^2) \| \mathcal{N}(0, I)) = \frac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$ regularizes the latent space. Reparameterization trick $z = \mu + \sigma \odot \epsilon, \epsilon \sim \mathcal{N}(0, I)$ enables gradient-based training.

Failure Mode: Encoder $\rightarrow$ Decoder $\rightarrow$ reconstruction loss = the model audits ITSELF. Theorem 2: q_ϕ and p_θ are jointly trained to minimize . The reconstruction term $\mathbb{E}_{q_\phi}[\log p_\theta(x|z)]$ uses the encoder’s z to evaluate the decoder’s $\hat{x}$. But q_ϕ was trained to make z easy to decode. This is collusion, not verification.

SCX Diagnosis: Blurry outputs are Theorem 3 in action. VAEs produce blurry images. Why? The reconstruction loss (typically MSE or binary cross-entropy) penalizes pixel-wise deviation. For uncertain z , the optimal reconstruction is the *expected* image $\mathbb{E}_{p_\theta(x|z)}[x]$, which is a weighted average of possible images $\rightarrow$ blur. But is the blur because the model is uncertain (good) or because the decoder is weak (bad)? Theorem 3: you cannot tell. One model cannot distinguish aleatoric uncertainty from epistemic deficiency.

Mathematical Audit: The VAE’s quality signal is:

$$\text{ReconstructionError} = \|x - \hat{x}\|^2 = \|x - \text{Decoder}(\text{Encoder}(x))\|^2.$$

Encoder and Decoder are trained jointly $\rightarrow \text{Cov}(\text{Encoder error}, \text{Decoder error}) > 0$. The effective audit multiplicity is $M_{\text{eff}} = 1/(1 + \rho_{\text{Enc-Dec}}) \approx 1$ since $\rho_{\text{Enc-Dec}} \approx 1$ for a jointly-trained autoencoder. Theorem 2: self-audit provides zero information about true generation quality.

What SCX Provides: External quality auditor: $M > 1$ independent discriminators evaluate VAE outputs against real data. YAJIE consensus across discriminators. SITUS latent space audit: $\text{Lip}(\text{Encoder}) \leq L_{\text{enc}}, \text{Lip}(\text{Decoder}) \leq L_{\text{dec}}$. CERCIS Score: $Q_{\text{VAE}} = 0.2$.

10.3 21. Diffusion Models Diffusion Models

Verdict: **PARTIAL** —Sequential “Audit” With Correlated Steps

Why Popular: SOTA image generation. Ho et al. (2020), Song et al. (2021): forward process $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$ adds Gaussian noise. Reverse process $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$ learns to denoise:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right).$$

The denoising network ϵ_θ predicts the noise at each timestep. DDPM (2020) uses $T = 1000$ steps; DDIM (2021) enables fewer steps. Stable Diffusion (2022) operates in latent space.

Failure Mode: Iterative denoising = sequential refinement by the SAME model. Each step t applies $\epsilon_\theta(x_t, t)$. The noise predictor is shared across all timesteps. $\epsilon_\theta(x_t, t)$ and $\epsilon_\theta(x_{t'}, t')$ are outputs of the same neural network with different time embeddings. The predictions are correlated:

$$(\epsilon_\theta(x_t, t), \epsilon_\theta(x_{t'}, t')) = \rho_{\text{arch}} > 0,$$

where ρ_{arch} is architectural correlation (shared weights). This is sequential dependency: $M_{\text{eff}} \ll T$.

SCX Diagnosis: Diffusion can be reframed as T sequential refinement steps. Theorem 1 requires independent experts. With T correlated applications of the same model: $M_{\text{eff}} = T/(1 + (T - 1)\rho_{\text{arch}}) \rightarrow 1/\rho_{\text{arch}}$ as $T \rightarrow \infty$. For $\rho_{\text{arch}} \approx 0.8$ (high, since same weights), $M_{\text{eff}} \approx 1.25$ —effectively one expert. The 1000 diffusion steps provide negligible audit benefit beyond ~ 2 independent assessments.

Mathematical Audit: The quality of a diffusion step at time t can be measured by the denoising error:

$$\varepsilon_t = \mathbb{E}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2].$$

This is the training loss. But this is self-audit: the model’s training objective evaluates itself (Theorem 2). The user never knows ε_t for a generated sample.

What SCX Provides: Independent auditors at each noise level: M different denoising networks $\epsilon_\theta^{(1)}, \dots, \epsilon_\theta^{(M)}$, each specialized to a noise range $[t_m^{\min}, t_m^{\max}]$. YAJIE consensus at each step. Or: fewer steps with $M > 1$ auditors per step, providing $\mathbb{P}(\text{miss}) \leq \exp(-2M\Delta^2)$ instead of pseudo-audit from T correlated model calls. CERCIS Score: $Q_{\text{Diffusion}} = 0.5$ (penalized for correlated steps).

11 Part V: Modern Paradigms —The Self-Audit Trap

12 Part V: Modern Paradigms — The Self-Audit Trap

Modern ML increasingly avoids human supervision. Self-supervised learning, transfer learning, meta-learning, reinforcement learning —each shifts the burden of quality assessment from humans to algorithms. This creates a structural audit crisis: when algorithms generate their own labels, evaluate their own outputs, and learn from their own predictions, who verifies the verifier?

12.1 22. Self-Supervised Learning Self-Supervised Learning — SimCLR/MoCo/MAE/BERT

Verdict: **UNDECLARED** —Self-Audit (Theorem 2)

Why Popular: No human labels needed. SSL generates supervisory signals from the data itself:

- **BERT** (Devlin et al., 2019): Mask 15% of tokens, predict masked tokens from context. Pre-train on BooksCorpus + Wikipedia.
- **SimCLR** (Chen et al., 2020): Apply two random augmentations to image x , encode both to z_i, z_j , maximize agreement between z_i and z_j while minimizing agreement with other images.
- **MoCo** (He et al., 2020): Maintain momentum encoder for consistent negative samples. Dictionary queue of encoded samples.
- **MAE** (He et al., 2022): Mask 75% of image patches, reconstruct from visible patches. Asymmetric encoder-decoder.

Failure Mode: Model generates its own labels → learns from self-generated labels → circular. Theorem 2: the mutual information between self-audit outcome and true quality is zero. BERT’s MLM accuracy measures how well BERT predicts BERT’s own masked tokens —not whether the learned representations encode genuine linguistic knowledge. The representations might encode dataset biases, superficial word co-occurrence, or artifacts of the masking pattern.

SCX Diagnosis: Exception: contrastive methods partially escape self-audit. Contrastive learning (SimCLR, MoCo) uses data augmentation to create multiple “views” of the same instance:

$$\text{contrastive} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}.$$

Different augmentations act as different “observers” of the same underlying instance $\rightarrow M_{\text{aug}} > 1$ pseudo-independent views. This is why SimCLR outperforms pure reconstruction SSL: it **accidentally** introduces $M > 1$, providing weak Theorem 1 benefits.

Mathematical Audit: The effective multiplicity of contrastive SSL with K augmentation types:

$$M_{\text{eff}} = \frac{K}{1 + \bar{\rho}_{\text{aug}}},$$

where $\bar{\rho}_{\text{aug}}$ is the average correlation between encodings from different augmentations. For SimCLR with random crop, color jitter, Gaussian blur, horizontal flip ($K = 4$), $\bar{\rho}_{\text{aug}} \approx 0.5 \rightarrow M_{\text{eff}} \approx 2.7$. Better than $M = 1$ but far from robust. Strong augmentations can destroy semantic content (rotated “6” “9”), creating false positives with no detection mechanism.

What SCX Provides: Explicit M declaration for augmentation diversity. Each augmentation type as noisy observation channel. YAJIE consensus across augmentation-specific encoders. CERCIS Score: $Q_{\text{SSL-recon}} = 0.1$ (pure reconstruction, self-audit); $Q_{\text{SSL-contrastive}} = 0.3$ ($M_{\text{aug}} > 1$ but unverified). SPRING permanently records which augmentations produce consistent vs. inconsistent representations.

12.2 23. Transfer Learning / Fine-tuning Transfer Learning / Fine-tuning

Verdict: **PARTIAL** —No Transfer Quality Metric

Why Popular: Works with limited data. Pre-train on massive dataset (ImageNet-21K, JFT-300M, C4), fine-tune on target task with few labeled examples. The pre-trained model θ_{pre} provides strong initialization; fine-tuning produces $\theta_{\text{ft}} = \theta_{\text{pre}} + \Delta\theta$ adapted to the target task. Dominant paradigm since 2018 —virtually all modern CV/NLP uses pre-trained backbones.

Failure Mode: **No metric for transfer quality. Negative transfer goes undetected.** Sometimes pre-training *hurts*: features learned on source data are irrelevant or harmful for the target. This is “negative transfer.” Current practice: try fine-tuning; if accuracy is good, use it; if not, train from scratch. No *a priori* method to predict transfer quality. No decomposition of final accuracy into “pre-trained contribution” vs. “fine-tuning contribution.”

SCX Diagnosis: Pre-training = state prior ρ_s estimation. Fine-tuning = Δ_s estimation. Theorem 3: cannot distinguish $\Delta_s \approx 0$ because (a) pre-training is genuinely useful, or (b) fine-tuning failed to correct pre-training’s errors. The two cases have identical final accuracy but radically different reliability.

Mathematical Audit: The transfer quality decomposition:

$$\text{Acc}_{\text{ft}} = \text{Acc}_{\text{scratch}} + \underbrace{(\text{Acc}_{\text{ft}} - \text{Acc}_{\text{scratch}})}_{\text{transfer gain } \Delta_{\text{transfer}}}.$$

$\Delta_{\text{transfer}} > 0$: positive transfer. $\Delta_{\text{transfer}} < 0$: negative transfer. But $\text{Acc}_{\text{scratch}}$ is unknown—you’d need to train from scratch to measure it, defeating the purpose of transfer learning. Estimating Δ_{transfer} without scratch training is an unidentifiability problem (Theorem 3).

What SCX Provides: CERCIS Score for transfer quality: $Q_{\text{transfer}} = \text{Acc}_{\text{ft}} - \text{Acc}_{\text{linear-probe}}$, where linear probe accuracy measures pre-trained feature quality independently of fine-tuning. SPRING detects negative transfer: records whether fine-tuning degrades linear probe performance. YAJIE consensus across M different pre-trained models verifies transfer robustness.

12.3 24. Few-Shot / Meta-Learning Few-Shot / Meta-Learning

Verdict: **UNDECLARED** — $M = K$, Data Scarcity vs. Audit

Why Popular: Mimics human rapid learning. Meta-learning learns to learn across tasks; at test time, adapts to new tasks from K examples:

- **MAML** (Finn et al., 2017): Learn initialization θ such that one gradient step on K examples produces good task-specific parameters.
- **Prototypical Networks** (Snell et al., 2017): Classify by distance to class prototypes $c_k = \frac{1}{K} \sum_i f_\phi(x_i)$.
- **Matching Networks** (Vinyals et al., 2016): Attention over support examples.

Failure Mode: K examples with no quality guarantee on any of them. Theorem 1: M is the number of independent experts. In few-shot learning, the effective audit multiplicity is at most $M_{\text{eff}} \leq K$, and typically much less due to within-class correlation. A single mislabeled support example poisons predictions for the entire class.

SCX Diagnosis: With $K = 1$ (one-shot learning): $M_{\text{eff}} \leq 1$. Theorem 1 collapse: no detection guarantee. With $K = 5$ (five-shot): $M_{\text{eff}} \leq 5$, but with within-class correlation $\bar{\rho} \approx 0.6$ (examples from same class share features), $M_{\text{eff}} \leq 5/1.6 \approx 3.1$. This is a **fundamental limitation**, not an implementation issue. Few-shot learning *cannot* have strong quality guarantees because the data itself provides insufficient independent verification.

Mathematical Audit: The error detection bound with K support examples:

$$\mathbb{P}(\text{miss}) \leq \exp(-2K_{\text{eff}}\Delta^2), \quad K_{\text{eff}} = \frac{K}{1 + \bar{\rho}_{\text{class}}}.$$

For $K = 5$, $\bar{\rho}_{\text{class}} = 0.6$, $\Delta = 1$: $\mathbb{P}(\text{miss}) \leq \exp(-2 \cdot 3.1) \approx 0.002$ —seems good! But this assumes each support example is an independent verifier, which is false: they all come

from the same meta-training distribution. The meta-training itself may have introduced biases that all K examples share ($\bar{\rho}_{\text{meta}} > 0$), further reducing K_{eff} .

What SCX Provides: Explicit M declaration: $M_{\text{eff}} \leq K/(1 + \bar{\rho}_{\text{class}} + \bar{\rho}_{\text{meta}})$. CERCIS Score explicitly caps $Q_{\text{few-shot}} \leq 1 - \exp(-2K_{\text{eff}}\Delta^2)$. YAJIE consensus across M different few-shot methods cross-validates predictions. SPRING permanently records support examples and their audit outcomes.

12.4 25. Reinforcement Learning Reinforcement Learning — DQN/PPO/SAC

Verdict: **UNDECLARED** — $M = 1$ **Reward Source**

Why Popular: Game-playing, robotics, sequential decision-making. RL maximizes cumulative reward:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right].$$

- **DQN** (Mnih et al., 2015): Deep Q-network with experience replay and target network.
- **PPO** (Schulman et al., 2017): $\max_{\theta} \mathbb{E}[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$.
- **SAC** (Haarnoja et al., 2018): $\max_{\pi} \mathbb{E}[\sum_t \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot|s_t)))]$.

AlphaGo/AlphaZero (Silver et al., 2016–2018) defeated world champions at Go, chess, shogi.

Failure Mode: **The environment is a single reward source.** $M = 1$. Theorem 1: no error detection. The environment provides one scalar reward per timestep. There is no second environment, no second reward function, no consensus on whether the reward accurately reflects true task performance. This explains **ALL major RL failure modes**.

SCX Diagnosis: **Reward hacking** is a Theorem 1 consequence: the agent finds behaviors that maximize reward but don't accomplish the intended task. A single reward function has loopholes. With $M > 1$ reward functions (different designers, different metrics), the agent must satisfy all simultaneously. The probability of finding a behavior that hacks all M reward functions decays exponentially in M .

Sample inefficiency: Each environment interaction provides one scalar signal. With M reward models providing M signals per interaction, the information per sample scales linearly with M . Sample efficiency is an audit problem: $M = 1$ yields one noisy signal; $M = 100$ yields a consensus signal with Hoeffding-guaranteed accuracy.

Mathematical Audit: For RL with $M = 1$ reward: $Q_1 = 0$ (no guarantee). For RL with $M > 1$ reward models:

$$\mathbb{P}(\text{all } M \text{ reward models fooled}) \leq \exp(-2M_{\text{eff}}\Delta_R^2),$$

where Δ_R is the reward gap between intended and hacked behavior. The SCX-augmented RL objective:

$$J_{\text{SCX}}(\pi) = \min_{m \in [M]} \mathbb{E}_{\tau \sim \pi} \left[\sum_t \gamma^t r_m(s_t, a_t) \right],$$

maximizing the **minimum** reward across all M reward models —conservative, safe, audited.

What SCX Provides: Multi-expert reward audit: $M > 1$ reward models from different designers/metrics. YAJIE consensus: policy satisfies all reward models. SPRING permanently records trajectories and reward model disagreements. CERCIS Score: $Q_{\text{RL}} = 0.1$ ($M=1$); $Q_{\text{SCX-RL}} = 1 - \exp(-2M\Delta_R^2)$ with $M > 1$.

12.5 26. Imitation Learning / Behavioral Cloning Imitation Learning / Behavioral Cloning

Verdict: **UNDECLARED** —Copying Without Audit

Why Popular: Simpler than RL. Learn from expert demonstrations $\mathcal{D}_E = \{(s_i, a_i)\}_{i=1}^N$:

$$\pi_{\text{BC}} = \arg \min_{\pi} \sum_{(s,a) \in \mathcal{D}_E} (\pi(s), a).$$

Dagger (Ross et al., 2011): iteratively collect more data by executing π and querying expert for correct actions at visited states. Used in autonomous driving (learn from human drivers), robotic manipulation (learn from teleoperation).

Failure Mode: **Expert demonstrations = single expert. No verification.** The demonstrator is one policy π_E — $M = 1$. If π_E makes mistakes, π_{BC} copies them. If π_E 's demonstrations don't cover all states, π_{BC} encounters distribution shift and fails catastrophically.

SCX Diagnosis: Distribution shift is a Theorem 1 consequence: the imitator visits states outside the expert's support. At those states, the imitator must extrapolate. Extrapolation error compounds —each step away from demonstrated states increases error. With $M = 1$, there is no consensus to detect off-distribution states. DAgger partially addresses this by collecting more expert data online, but still has a single expert.

Mathematical Audit: The error bound for behavioral cloning under distribution shift:

$$\mathbb{E}_{s \sim d_{\pi}} [(\pi(s), \pi^*(s))] \leq \varepsilon + O\left(\frac{1}{1-\gamma} \text{TV}(d_{\pi} \| d_{\pi_E})\right),$$

where d_{π} is state visitation distribution of learned policy, d_{π_E} is expert's distribution. As $\text{TV}(d_{\pi} \| d_{\pi_E})$ grows (distribution shift), error grows unboundedly. With $M > 1$ experts, $\text{TV}(d_{\pi} \| d_{\pi_E})$ can be bounded by consensus across experts' state distributions.

What SCX Provides: Multi-expert demonstration audit: learn from $M > 1$ experts with different styles. YAJIE consensus across expert policies identifies ambiguous states (experts disagree). SPRING records which expert was followed at each state. CERCIS Score: $Q_{BC} = 0.1$ ($M=1$); $Q_{SCX-BC} = 1 - \exp(-2M_E\Delta^2)$.

12.6 27. Recommendation Systems / Collaborative Filtering Recommendation Systems

Verdict: **UNDECLARED** —Self-Audit via Click-Through Rate

Why Popular: Powers internet commerce. Matrix factorization: $R \approx U^T V$ where U is user embeddings, V is item embeddings. Deep recommender systems (Wide & Deep, DLRM, DCN) combine collaborative filtering with content features. Optimize for click-through rate (CTR), conversion rate, watch time.

Failure Mode: **CTR is a self-audit metric.** The system recommends items; users click (or not); clicks validate the recommendation. But: recommendations influence what users see $\rightarrow$ clicks are biased toward recommended items $\rightarrow$ the system reinforces its own biases. Theorem 2: self-generated feedback (clicks on system-recommended items) is self-audit. The true quality —would the user have preferred an item they never saw? —is fundamentally unmeasurable within the system.

SCX Diagnosis: The feedback loop recommend $\rightarrow$ click $\rightarrow$ retrain $\rightarrow$ recommend is a self-audit cycle. The system can achieve high CTR by recommending popular items that would be clicked regardless —this is the “popularity bias” that degrades discovery. Theorem 2: the system’s CTR tells you about the system’s ability to predict clicks on its own recommendations, not about recommendation quality.

What SCX Provides: $M > 1$ independent recommendation engines with different architectures. YAJIE consensus on recommendations: items recommended by all M engines are verified. SPRING permanently records (user, recommended items, clicks, non-clicks) for retrospective audit. CERCIS Score: $Q_{RecSys} = 0.15$ (self-audit feedback loop).

13 Part VI: The Verdict Part VI: The Verdict

We have audited 27 algorithms across 70 years of ML. The results are concerning. Only 5 achieve **VERIFIED** status, and even those leave critical gaps. The rest range from **PARTIAL** (structural limitations) to **UNDECLARED** (never considered audit at all). This is not a close call. This is a systemic, field-wide failure of epistemic hygiene.

13.1 Summary of Convictions —One Line Per Algorithm Summary of Convictions

- **Linear / Logistic Regression:** $M = 1$. 70 years, never audited. **UNDECLARED**
- **k-Nearest Neighbors:** State encoding without quality metric. **UNDECLARED**
- **Naive Bayes:** Assumes independence, never verifies. **UNDECLARED**
- **Decision Trees:** One tree, one opinion. Overfits silently. **UNDECLARED**
- **SVM:** Kernel SITUS without Lipschitz bound. **UNDECLARED**
- **k-Means:** Arbitrary K, no cluster quality guarantee. **UNDECLARED**
- **Bagging:** Good structure, no M_{eff} declaration. **PARTIAL**
- **Random Forest:** Best classical algorithm. Accidentally implements Theorem 1. **VERIFIED**
- **XGBoost:** Sequential dependency breaks independence. **UNDECLARED**
- **Stacking:** Meta-learner introduces new unverified component. **PARTIAL**
- **Deep MLPs:** Depth without purpose. Theorem 3 violation. **UNDECLARED**
- **CNN:** SITUS without Lipschitz. Max-pooling destroys information. **PARTIAL**
- **RNN/LSTM:** Forget gate = evidence destruction. SPRING violation. **UNDECLARED**
- **ResNet:** Signal-noise separation. Best deep architecture. **VERIFIED**
- **Batch Normalization:** Satisfies A5. Breaks at small batch. **VERIFIED**
- **Dropout:** 2^n sub-networks voting. Theorem 1 implicit. **VERIFIED**
- **Transformer:** Multi-head = M queries. KV cache = bounded SPRING. **VERIFIED**
- **BERT:** Self-supervised $\rightarrow$ self-audit (Theorem 2). **UNDECLARED**
- **GPT:** Autoregressive $M = 1$. Hallucinations undetectable. **UNDECLARED**
- **GAN:** $M = 1$ discriminator $\rightarrow$ mode collapse, instability. **UNDECLARED**
- **VAE:** Encoder-Decoder collusion = self-audit. **UNDECLARED**
- **Diffusion Models:** Correlated denoising steps. $M_{\text{eff}} \approx 1$. **PARTIAL**
- **Contrastive SSL:** $M_{\text{aug}} > 1$ but unverified. **UNDECLARED**
- **Reconstruction SSL:** Pure self-audit. $M = 0$. **UNDECLARED**
- **Transfer Learning:** No transfer quality metric. **PARTIAL**
- **Meta-Learning:** $M \leq K$ with correlation. Fundamental audit limit. **UNDECLARED**
- **RL:** $M = 1$ reward. Reward hacking, sample inefficiency. **UNDECLARED**
- **Imitation Learning:** $M = 1$ expert. Distribution shift silently compounds. **UNDECLARED**
- **Recommendation Systems:** CTR feedback loop = self-audit. **UNDECLARED**

Statistics: 29 algorithms audited. 5 verified. 6 weaknesses. 18 undeclared. **Verdict rate: 62.1% audit finding.**

13.2 The Cercis Score Ranking —All Algorithms Cercis Score Ranking

Algorithm	Q	N	S	Verdict
Random Forest (Random Forest)	0.90	0.20	0.94	VERIFIED Near-SCX
ResNet (ResNet)	0.85	0.40	0.93	VERIFIED
Transformer (Atten- tion)	0.80	0.50	0.90	VERIFIED
Dropout	0.80	0.30	0.86	VERIFIED
BatchNorm (Batch Normalization)	0.70	0.10	0.72	VERIFIED
Bagging (Bagging)	0.60	0.10	0.62	PARTIAL
CNN (CNN)	0.55	0.45	0.64	PARTIAL
Diffusion (Diffusion Models)	0.50	0.60	0.62	PARTIAL
Stacking (Stacking)	0.45	0.10	0.47	PARTIAL
Transfer Learning (Transfer Learning)	0.40	0.40	0.48	PARTIAL
XGBoost	0.50	0.20	0.54	UNDECLARED Sequential
SVM (SVM)	0.40	0.30	0.46	UNDECLARED No Lipschitz
Naive Bayes (Naive Bayes)	0.35	0.10	0.37	UNDECLARED Unverified In- dep
Decision Tree (Deci- sion Trees)	0.30	0.10	0.32	UNDECLARED M=1

Continued

Algorithm	Q	N	S	Verdict
Deep MLP (Deep MLP)	0.30	0.25	0.35	UNDECLARED Depth w/o Purpose
SSL —Contrastive (Contrastive SSL)	0.30	0.70	0.44	UNDECLARED Weak M
Linear Reg. (Linear Regression)	0.25	0.10	0.27	UNDECLARED M=1
k-NN (k-Nearest Neighbors)	0.20	0.10	0.22	UNDECLARED Lazy Audit
k-Means (k-Means)	0.20	0.10	0.22	UNDECLARED Arbitrary K
SSL —Recon. (Reconstruction SSL)	0.15	0.60	0.27	UNDECLARED Self-Audit
VAE (VAE)	0.20	0.45	0.29	UNDECLARED Self-Audit
RNN/LSTM (RNN)	0.15	0.30	0.21	UNDECLARED Memory De- cay
Meta-Learning (Meta-Learning)	0.15	0.35	0.22	UNDECLARED M=K
RecSys (Recommendation Systems)	0.15	0.50	0.25	UNDECLARED Self-Audit
GAN (GAN)	0.10	0.90	0.28	UNDECLARED M=1 Dis- criminator
RL / PPO (RL)	0.10	0.80	0.26	UNDECLARED M=1 Reward

Continued

Algorithm	Q	N	S	Verdict
Imitation Learning (Imitation Learning)	0.10	0.40	0.18	UNDECLARED M=1 Expert

表 1: CERCIS Score ranking. Q : formal guarantee (0–1). N : empirical novelty (0–1). $\eta = 0.2$. $S = Q + \eta N$. Algorithms with $Q < 0.3$ are epistemically worthless regardless of empirical performance.

13.3 Analysis of Rankings Analysis of Rankings

Tier 1 — VERIFIED VERIFIED ($S > 0.70$, **VERIFIED**): Random Forest ($S = 0.94$), ResNet ($S = 0.93$), Transformer ($S = 0.90$), Dropout ($S = 0.86$), BatchNorm ($S = 0.72$). These algorithms accidentally implement core SCX theorems through their architectural design:

- **Random Forest**: M explicit trees + bootstrap (A1) + random subspaces (A2) + OOB error (cross-audit). The best-designed algorithm in ML history —and even it lacks explicit audit declaration.
- **ResNet**: $h_{\ell+1} = h_{\ell} + F_{\ell}(h_{\ell})$ = explicit signal-noise separation. The only architecture that formally enables Theorem 3 audit. Missing explicit per-block verification.
- **Transformer**: Multi-head = M parallel SPRING gatekeepers. Softmax attention = probabilistic memory access. Missing unbounded SPRING memory —bounded KV cache limits audit horizon.
- **Dropout**: 2^n sub-networks voting at test time. MC Dropout provides uncertainty estimates —but these are self-audited (Theorem 2) unless cross-validated with external metrics.
- **BatchNorm**: Enforces A5 (bounded state distribution). Breaks at small batch sizes —the estimator becomes the bottleneck.

Tier 2 — PARTIAL PARTIAL ($0.45 \leq S \leq 0.70$, **PARTIAL**): Bagging, CNN, Diffusion, Stacking, Transfer Learning. These have audit structure but critical gaps: Bagging has bootstrap independence but no M_{eff} correction; CNN has spatial SITUS but no Lipschitz guarantee; Diffusion has sequential refinement but correlated steps; Stacking learns combination weights but the meta-learner is unverified; Transfer Learning reuses features but can’t measure transfer quality.

Tier 3 — UNDECLARED UNDECLARED ($S < 0.55$, **UNDECLARED**): The remaining 17 algorithms. Failure modes cluster into five categories:

Failure 1: $M = 1$ ($M = 1$): Linear Regression, Decision Tree, GAN (single discriminator), RL (single reward), Imitation Learning (single expert). Single point of failure —

Theorem 1 provides no guarantee.

- Failure 2: Self-Audit (Self-Audit, Theorem 2):** VAE (encoder evaluates decoder), SSL-reconstruction (self-generated labels), RecSys (CTR feedback loop). The system grades its own homework —epistemically equivalent to no grading.
- Failure 3: Sequential Dependency (Sequential Dependency):** XGBoost/Boosting (each tree depends on previous errors), RNN/LSTM (hidden state depends on all previous), Diffusion (correlated denoising steps). $M_{\text{eff}} \ll M$, breaking A1/A2.
- Failure 4: Unverified Structure (Unverified Structure):** SVM (kernel Lipschitz uncontrolled), k-Means (K chosen heuristically), Naive Bayes (independence assumed, never checked), k-NN (state encoding without quality metric), Deep MLP (depth unverified).
- Failure 5: Data Scarcity (Data Scarcity):** Meta-Learning (M=K with high correlation). Fundamental limit: few-shot learning with K examples cannot exceed quality guarantee determined by K.

13.4 What Every Algorithm Lacks —The SCX Gap What Every Algorithm Lacks

Every algorithm audited, including the **VERIFIED** ones, lacks the following seven SCX components. This is the audit gap —the structural distance between current ML and verifiable ML.

1. **M parameter declaration (M Parameter Declaration).** No algorithm declares M , the minimum number of independent experts needed to certify output quality at ε . Without M , the user cannot assess verifiability. M is the most fundamental epistemic parameter in any learning system. Its absence is not a gap —it’s a void.
2. **Independent multi-expert consensus —Yajie (Independent Multi-Expert Consensus).** Algorithms with multiple components have implicit experts, but none implement formal YAJIE Nash equilibrium consensus with provable error detection bounds. Majority voting, weighted averaging, and learned combination are heuristics —they work empirically but provide no formal guarantee.
3. **Spring permanent, monotonic memory (Spring Permanent Monotonic Memory).** No algorithm maintains permanent, non-forgetting memory. LSTMs have forget gates. Transformers have bounded context windows. RL agents discard old trajectories. Diffusion overwrites intermediate samples. Every algorithm forgets —and therefore every algorithm loses audit evidence.
4. **Cercis Score quality metric (Cercis Score).** No algorithm provides $S = Q + \eta N$ as a single auditable quality number. Users rely on benchmark accuracy (proxy for N) and ignore Q (formal guarantee). A model with 99% accuracy and $Q = 0$ is epistemically worthless —but users can’t tell because Q is never reported.

5. **Theorem 3 unidentifiability awareness (Theorem 3 Unidentifiability Awareness).** No algorithm acknowledges that added depth, parameters, or training data cannot distinguish signal from noise without independent audit. Architectures add layers and parameters with no justification beyond improved training loss —self-audit of the worst kind.
6. **Cryptographic hash binding —Cryptographic Hash Binding.** No algorithm provides SHA-256 commitment to (model weights || training data manifest || hyperparameters || declared M). Users download weights with no guarantee that the running model matches the audited model. Audit without cryptographic binding is audit without evidence.
7. **Distribution shift detection —Spring gating (Distribution Shift Detection).** No algorithm actively monitors whether its input distribution has shifted from training. Models degrade silently. SPRING provides explicit gating: when $\text{KL}(\rho_t || \rho_{\text{train}}) > \tau$, the model is flagged as unreliable. This is absent from every existing algorithm.

13.5 The Audit Taxonomy —Three Eras, One Failure

The 70-year history of ML is a progression through three eras, each with a characteristic audit failure. But the underlying pattern is constant: optimize prediction quality, ignore verification quality.

Era 1: Classical ML (1958–2000) — M Not Considered. The concept of audit didn’t exist. Algorithms were designed to predict, not verify. Linear regression, decision trees, SVMs, k-means —all $M = 1$. The failure mode is **absence**: audit structure is missing entirely because verification wasn’t recognized as a problem.

Era 2: Ensemble ML (1995–2015) — M Discovered Empirically. Bagging, boosting, random forests —practitioners discovered multiple models outperform single models. But the discovery was empirical, not theoretical. They didn’t know *why* ensembles work, so they couldn’t design for audit. Bagging uses bootstrap (good for independence) but doesn’t measure correlation. Boosting builds sequential models (destroys independence) but doesn’t realize this is harmful. The failure mode is **approximation**: the structure is partially present but unformalized.

Era 3: Deep Learning (2012–Present) —Depth as Compensatory Mechanism. Deep neural networks stack layers to improve performance. Theorem 3 reveals the dark possibility: depth compensates for noise, not just learns structure. ResNet separates signal from noise; Transformer provides multi-head parallelism —these are bright spots. But most deep architectures use depth without audit. The failure mode is **complexity without verification**: more layers, more parameters, more data —all with no way to tell if they help or hurt verifiability.

13.6 Complete Verdict Table Complete Verdict Table

14 Epilogue —The Path Forward Epilogue — The Path Forward

SCX is not “another algorithm.” It is not a competitor to Random Forest, Transformer, or ResNet. It is the **audit layer** —the verification infrastructure that every algorithm should have had from the beginning. The 70-year history of ML is the history of algorithms that work —until they don’t —with no way to know when or why.

14.1 The Fundamental Insight The Fundamental Insight

Every ML algorithm produces a prediction $\hat{y} = f(x; \theta)$. A user deploying this prediction in a safety-critical, financial, or medical context wants to know: **is this specific prediction reliable?**

The current answer is always some form of self-audit: test set accuracy (data the model already saw), confidence score (generated by the model itself), benchmark ranking (aggregate statistics that hide individual failures). None of these answers the question.

SCX answers differently: **declare** M . How many independent experts would need to produce the same prediction for you to trust it? If M is high and verified, trust. If M is low or undeclared, consider the prediction unverified.

This is not a novel concept. Peer review in science requires independent reviewers. Jury trials require multiple jurors. Engineering safety requires independent verification and validation. Financial auditing requires external accountants. Every domain with high-stakes decisions has discovered, through painful experience, that **no single observer is reliable**. ML is the only field that hasn’t learned this lesson.

14.2 The SCX Prescription —Required Components

For any ML algorithm to achieve SCX certification, it must implement:

Requirement 1: Declare M (Declare M). For target confidence $1 - \varepsilon$ and error magnitude Δ , declare $M \geq \ln(1/\varepsilon)/(2\Delta^2) \cdot (1 + \bar{\rho})$. Default: $\varepsilon = 0.05, \Delta = 0.5, \bar{\rho} = 0.2 \Rightarrow M \geq 8$. If the algorithm inherently has $M = 1$, it is non-compliant and must be wrapped in a multi-expert architecture.

Requirement 2: Implement Yajie Nash consensus (Implement Yajie Nash Consensus). M independent experts with verifiable competence scores ω_m . The consensus $\hat{y}_{\text{YAJIE}} = \text{weighted_median}(\hat{y}_1, \dots, \hat{y}_M; \omega_1, \dots, \omega_M)$ must be a Nash equilibrium: no coalition of experts can improve the aggregate prediction by deviating from honest reporting. The consensus mechanism itself is a fixed mathematical function —no learned parameters.

Requirement 3: Deploy Spring permanent memory (Deploy Spring Permanent Memory).

Every prediction, outcome, and audit result stored permanently. $M_t = M_{t-1} \cup \{(x_t, \hat{y}_t, y_t, \text{audit_verdict}_t)\}$. Memory is monotonic non-decreasing. No forgetting. No bounded capacity. No context window limits. The audit trail is complete and immutable forever.

Requirement 4: Compute and publish Cercis Score (Compute and Publish Cercis Score).

$S = Q + \eta N$ with every model release. Q decomposes into theorem-specific scores: $Q = \frac{1}{5}[q_1(\text{Thm 1}) + q_2(\text{Thm 2}) + q_3(\text{Thm 3}) + q_{\text{SITUS}}(\text{Situs-1}) + q_{\text{SPRING}}(\text{Spring-1})]$. Users can compare models by S , knowing that $S < 0.3$ means the model is epistemically worthless.

Requirement 5: Acknowledge Theorem 3 (Acknowledge Theorem 3).

Every architectural decision —depth, width, skip connections, normalization, attention heads —must be justified against a noise audit baseline. If a component cannot be shown to reduce verifiable error (not just training error), it is noise compensation and must be flagged.

Requirement 6: Bind with cryptographic hash (Cryptographic Hash Binding).

SHA-256 hash $H = \text{hash}(\text{weights} \parallel \text{data_manifest} \parallel \text{hyperparameters} \parallel M \parallel \text{Cercis Score})$. The hash is published immutably. Any change to the model, data, or quality claims changes the hash —tampering is detectable. The binding transforms M from a **claim** into a **commitment**.

Requirement 7: Monitor distribution shift via Spring gating (Spring Gating Monitoring).

At test time, compute $\text{KL}(\rho_{\text{test}} \parallel \rho_{\text{train}})$ for the input distribution. When KL divergence exceeds threshold τ , the model's predictions are flagged as unverified due to distribution shift. This is not a model limitation —it is an honest declaration of the model's applicable domain.

14.3 The Arms Race We Need The Arms Race We Need

M-declaration creates a virtuous arms race. The mechanism:

1. Company A releases Model-A with declared $M = 50$, measured $\bar{\rho} = 0.15$, $\text{CERCIS}(A) = 0.90$.
2. Company B releases Model-B with undeclared M —effectively $M = 0$.
3. Regulators, customers, and users compare: Model-A is verified; Model-B is not.
4. Company B is forced to declare M or lose market share.
5. Both companies increase M , driving quality guarantees upward.

This is not speculative. It is the same dynamic that drove safety ratings in automobiles, energy efficiency labels in appliances, and nutritional labels in food. **Information asymmetry favors the seller; mandatory disclosure levels the field.** M-declaration is mandatory disclosure for ML quality.

14.4 The Dark Forest Protocol The Dark Forest Protocol

Any entity publishing non-SCX M parameters broadcasts “dark forest coordinates”:

- **Non-SCX M = self-exposure.** Self-developed audit without the mathematical guarantees of Theorems 1–5. By Theorem 2, non-SCX M lacks validity.
- **Falsification triggers permanent record.** SCX-certified M with $M_{\text{true}} < M_{\text{declared}}$ is active fraud (Theorem 3 from the Science Audit). $\mathbb{P}(\text{detection}) \rightarrow 1$ as SPRING accumulates evidence.
- **Non-participation tolerated but flagged.** Entities not publishing M are $M = 0$ by default. SCX does not pursue —they are irrelevant until entering the verification game.

14.5 Final Words Final Words

No M , No Trust.

Without M , no trust.

$M = 1$ is epistemically equivalent to $M = 0$.

$M=1$ is epistemically equivalent to $M=0$.

Self-audit is no audit.

Self-Audit = No Audit.

Sequential experts are one expert.

Sequentially dependent experts are equivalent to one expert.

The 70-year detour ends now.

Seventy years of detours, ending here.

Declare M . Verify. Trust nothing else.

Declare M . Verify. Beyond this, there is no trust.

—SCX, June 2026

14.6 Appendix: Quick Reference —SCX Audit of All Algorithms Appendix: Quick Reference

How to read this appendix: Each entry lists the algorithm, its verdict, the SCX theorem it violates, its M_{eff} (effective multiplicity), and the primary missing SCX component.

This serves as a one-page reference for researchers evaluating which algorithms to trust —and which to replace.

Algorithm	Verdict M_{eff}	Violates	Missing SCX
Linear Regression	1 UNDECLARED	Thm 1	YAJIE consensus
k-Nearest Neighbors	$\leq k/2$ UNDECLARED	Situs-1	SITUS Lipschitz
Naive Bayes	1 UNDECLARED	Thm 2, A2	Feature audit
Decision Tree	1 UNDECLARED	Thm 1, Thm 2	Multi-tree consensus
SVM	1 UNDECLARED	Situs-1	Kernel certification
k-Means	1 UNDECLARED	Thm 2	CERCIS cluster score
Bagging	$B/(1+\rho)$ PARTIAL	—(undeclared)	M_{eff} declaration
Random Forest	$B/(1+\rho)$ VERIFIED	—(undeclared)	CERCIS + SPRING
XGBoost	$\ll T$ UNDECLARED	A1 (sequential)	Parallel experts
Stacking	M PARTIAL (base), 1 (meta)	Thm 1 (meta)	Meta-learner audit
Deep MLP	1 UNDECLARED	Thm 3	Depth audit
CNN	1 PARTIAL	Situs-1	Lipschitz bound
RNN/LSTM	1 (sequential) UNDECLARED	Spring-1	Permanent memory
ResNet	1 (but separable) VERIFIED	—(undeclared)	Per-block audit
BatchNorm	1 VERIFIED	—(A5 satisfied)	Drift monitor

Continued

Algorithm	Verdict M_{eff}	Violates	Missing SCX
Dropout	$\ll 2^n$ VERIFIED	—(undeclared)	M_{eff} declaration
Transformer	$h/(1+\bar{\rho})$ VERIFIED	Spring-1 (bounded KV)	Unbounded memory
GAN	1 (discriminator) UNDECLARED	Thm 1	$M > 1$ discriminators
VAE	≈ 1 UNDECLARED	Thm 2	External auditor
Diffusion	$\approx 1/\rho$ PARTIAL	A1 (steps correlated)	Independent step auditors
SSL (contrastive)	$K/(1+\bar{\rho})$ UNDECLARED	Thm 2 (partial)	Augmentation audit
SSL (reconstruction)	0 UNDECLARED	Thm 2	External labels
Transfer Learning	1 PARTIAL	Thm 3	Transfer quality metric
Meta-Learning	$\leq K/(1+\bar{\rho})$ UNDECLARED	Thm 1 (hard limit)	Support set audit
RL (PPO/DQN)	1 (reward) UNDECLARED	Thm 1	$M > 1$ reward models
Imitation Learning	1 (expert) UNDECLARED	Thm 1	Multi-expert audit
Recommendation	≈ 1 UNDECLARED	Thm 2	External feedback

表 3: Quick Reference: SCX Audit Summary. M_{eff} = effective multiplicity. “Violates” = which theorem/assumption is structurally violated. “Missing SCX” = the primary SCX component that would fix the deficiency.

Interpretation: Algorithms with $M_{\text{eff}} = 1$ are structurally unverifiable. No amount of benchmark accuracy can fix this. Algorithms with $M_{\text{eff}} > 1$ but undeclared have audit structure but lack formal certification. Only algorithms with declared M_{eff} and explicit CERCIS Scores are SCX-compliant. Today, **zero** algorithms meet this standard.

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表 2: Complete Audit Verdicts — 27 Algorithms, 27 Convictions

Algorithm	Verdict	§	Core Deficiency (Core Deficiency)
Linear Regression (Linear Regression)	UNDECLARED	3.1	$M = 1$, single weight vector, no error detection
k-Nearest Neighbors (k-Nearest Neighbors)	UNDECLARED	3.2	Lazy state encoding, no SITUS Lipschitz bound
Naive Bayes (Naive Bayes)	UNDECLARED	3.3	Unverified independence assumption on features
Decision Tree (Decision Trees)	UNDECLARED	3.4	$M = 1$, overfits silently, self-audit leaf probs
SVM (SVM)	UNDECLARED	3.5	Kernel SITUS without Lipschitz, no reconstruction
k-Means (k-Means)	UNDECLARED	3.6	Arbitrary K, self-referential WCSS metric
Bagging (Bagging)	PARTIAL	4.1	Bootstrap independence but no M_{eff} declaration
Random Forest (Random Forest)	VERIFIED	4.2	Best classical; lacks explicit Cercis + Spring
XGBoost/Boosting	UNDECLARED	4.3	Sequential experts violate A1, $M_{\text{eff}} \ll M$
Stacking (Stacking)	PARTIAL	4.4	Meta-learner $M = 1$, unaudited weight learning
Deep MLP (Deep MLPs)	UNDECLARED	5.1	Depth without Theorem 3 audit, vanishing gradients
CNN (CNN)	PARTIAL	5.2	SITUS spatial encoding without Lipschitz guarantee
RNN/LSTM (RNN)	UNDECLARED	5.3	Forget gate destroys evidence (Spring-1 violation)
ResNet (ResNet)	VERIFIED	5.4	Best deep arch; missing per-block verification
BatchNorm (Batch Normalization)	VERIFIED	5.5	Satisfies A5; breaks at small batch sizes
Dropout	VERIFIED	5.6	2^n sub-networks voting; M_{eff} undeclared
Transformer (Attention)	VERIFIED	5.7	Multi-head = M queries; KV cache bounded memory
BERT (BERT)	UNDECLARED	5.8	Self-supervised $\rightarrow$ self-audit (Theorem 2)
GPT (GPT)	UNDECLARED	5.8	Autoregressive $M = 1$, hallucinations undetectable
GAN (GAN)	UNDECLARED	6.1	$M = 1$ discriminator $\rightarrow$ mode collapse, instability
VAE (VAE)	UNDECLARED	6.2	Encoder-Decoder collusion = self-audit (Theorem 2)
Diffusion (Diffusion Models)	PARTIAL	6.3	Correlated denoising steps, $M_{\text{eff}} \approx 1$
SSL —Contrastive (Contrastive SSL)	UNDECLARED	7.1	$M_{\text{aug}} > 1$ but unverified, augmentation destroys semantics
SSL —Reconstruction (Reconstruction SSL)	UNDECLARED	7.1	Pure self-audit, self-generated labels (Theorem 2)
Transfer Learning (Transfer Learning)	PARTIAL	7.2	No transfer quality metric, negative transfer undetected
Meta-Learning (Meta-Learning)	UNDECLARED	7.3	$M \leq K$ with class correlation, fundamental audit limit
RL / PPO (RL)	UNDECLARED	7.4	$M = 1$ reward, reward hacking, sample inefficiency
Imitation Learning (Imitation Learning)	UNDECLARED	7.5	$M = 1$ expert, distribution shift compounds silently
RecSys (Recommendation Systems)	UNDECLARED	7.6	CTR feedback loop = self-audit (Theorem 2)